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Delayed Weight Updates for High Throughput Hardware Accelerator of the CCSDS 123.0-B-2 Compression Algorithm

Master’s thesis in Electronic System Design and Innovation
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Abstract

Sammendrag

AI Declaration

Contents

[1] The section on hardware accelerators is taken as is from project thesis (TODO: make proper cite)

Acronyms

ADC Analog-to-Digital Converter. 15

ASIC Application Specific Integrated Circuit. 14, 15

AXI Advanced eXtensible Interface. 20, 21

BI Band Interleaved. 3

BIL Band Interleaved by Line. 3, 4, 11, 20, 21

BIP Band Interleaved by Pixel. 3, 4, 11, 20

BRAM Block Random Access Memory. 14, 15

BSQ Band Sequential. 3, 11

CCSDS Consultative Committee for Space Data Systems. 2

CLB Configurable Logic Block. 14

COTS Component off-the-Shelf. 15

CPU Central Processing Unit. 14, 15, 23, 24, 31

DMA Direct Memory Access. 15

DPCM Differential Pulse Code Modulation. 2

DSP Digital Signal Processor. 14, 15

FF Flip-Flop. 14, 15

FIFO First-In-First-Out. 21

FL Fast Lossless. 2

FLEX Fast Lossless Extended. 2

FPGA Field-Programmable Gate Array. 1, 2, 14, 15

GPO2 Golomb power-of-two. 13

GPU Graphical Processing Unit. 14, 15

HLS High-Level Synthesis. 15

HYPSO HYPerspectral Smallsat for Ocean observation. 1, 2, 15, 19, 20, 23

LSB Least Significant Bit. 22

LUT Look-Up-Table. 14, 15

MPSoC Multiprocessor System-on-Chip. 15, 20

MSB Most Significant Bit. 22

MSE Mean Square Error. 6

NTNU Norwegian University of Science and Technology. 1

PE Processing Element. 14

PL Programmable Logic. 15

PnR Place and Route. 15, 19

RTL Register Transfer Level. 15, 19

V2V Variable-to-Variable. 12, 13

Chapter 1

Introduction

With the development of small satellite technologies such as cube satellites (cubesats), with lower costs and shorter development times, the availability of remote sensing data has increased. Hereunder, hyperspectral and multispectral imagers provide data that is vital for ocean monitoring, geological research and observation of other natural processes [1, 2].

As part of the HYPerspectral Smallsat for Ocean observation (HYPSO) project at the Norwegian University of Science and Technology (NTNU), two 6U cubesats, HYPSO-1 and HYPSO-2, were launched in 2022 and 2024 respectively. These carry hyperspectral imagers capable of sensing wavelengths ranging from 430 nm to 800 nm over 120 spectral bands. Their primary purpose is to monitor algal blooms in the ocean, which can give indications as to the health of marine environments [3]. This is made possible by the reflection of characteristic wavelengths from chlorophyll, detectable by the on-board hyperspectral imagers.

A nominal HYPSO image has a size of roughly 150 MB. This presents a challenge due to limited on-board resources for storing large amounts of data, and low downlink speeds. As an example, HYPSO-1 is restricted to 75 MB of downlinked data per orbital revolution [3]. This prompts the need for on-board image compression to increase the scientific output and value of the mission.

Small satellites operate under strict limitations on on-board resources such as processor time and power budgets. This, combined with the complexity of image compression algorithms, makes the use of custom hardware accelerators ideal. For their implementation, Field-Programmable Gate Array (FPGA)s are the perfect choice, providing great flexibility and speed at a low cost.

This thesis explores the use of a novel weight update scheme proposed by Jia et al. for the CCSDS-123.0-B-2 compression algorithm. The purpose is to increase the throughput of the accelerator implemented by [5] for use on-board the HYPSO missions. This should not come at significant degradation in compression performance or hardware resource utilization. Additionally, an open-source decompressor has been implemented in order to verify the results of the proposed change in the algorithm.

The rest of the paper is organized as follows...

Chapter 2

Background

2.1 Hyperspectral Image Compression

The Consultative Committee for Space Data Systems (CCSDS) is a multi-institutional working group developing algorithms and standards for increased cooperation amongst space actors. A portion of their work has been concerned with the effective encoding of hyperspectral and multispectral images. Standard CCSDS 123.0-B-1, hereby referred to as issue 1, describes a lossless compression algorithm tailored for high throughput FPGA implementations. Issue 1 has been implemented and used successfully by several missions, the HYPSO project being one of them [1, 6]. In 2019, Issue 1 was followed by standard CCSDS 123.0-B-2, referred to as Issue 2. The following sections are based on the description of issue 2 presented in The Consultative Committee for Space Data Systems, 'Low-Complexity Lossless and Near-Lossless Multispectral and Hyperspectral Image Compression,' CCSDS, Recommended Standard CCSDS 123.0-B-2, 2019.

Issue 1 formalizes the Fast Lossless (FL) algorithm presented by Klimesh, which uses an adaptive filtering prediction technique to achieve lossless compression at low computational complexity suitable for space applications [8, 9]. To meet the needs of limited satellite downlink capabilities, lossy compression techniques have seen increasing use [9]. For this reason, issue 2 introduces a new near-lossless feature based on the Fast Lossless Extended (FLEX) algorithm presented by Keymeulen et al. [10]. Near-lossless compression, as opposed to lossy compression, provides the user with fine-tuned control over the per-pixel loss through tunable error limit parameters.

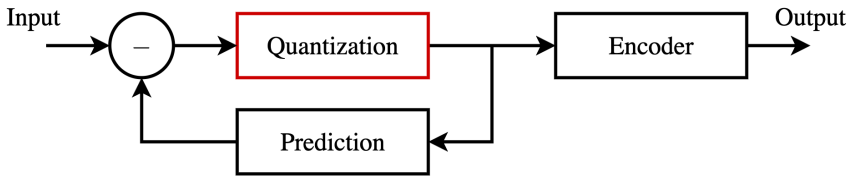


Figure 2.1: Layout of a differential pulse code modulator. The prediction loop, with optional quantization, reduces the entropy in the data passed to the encoder.

The issue 1 and issue 2 compressors use a form of Differential Pulse Code Modulation (DPCM), presented in Figure 2.1. A feedback loop, with optional quantization, predicts the value of the current input sample based on previously processed samples. The difference between the sample and the prediction, called the prediction residual, is encoded in the final bitstream. This exploits the correlation between neighbouring samples inherent in image data to reduce the entropy of the data passed to the encoder, allowing a more efficient encoding. Samples are processed in a single pass using simple operations. This reduces memory requirements and makes the algorithm suitable for hardware implementations. For the rest of this document, the prediction feedback loop will be referred to as

the predictor.

Predictor outputs are passed to an entropy coder which losslessly encodes prediction residuals in the final bitstream. Entropy coders, which will be covered in greater detail later in this document, encode samples with a varying number of bits based on the frequency of their occurrence in the image. A novel feature of issue 2 is the inclusion of a hybrid encoder, in addition to the sample-adaptive and block-adaptive encoders of issue 1. This performs better at the low encoding bitrates achievable in the near-lossless mode. After encoding, a header containing image metadata and compressor configuration parameters is prepended to the compressed bitstream (image body). This is required by the decompressor in order to correctly reconstruct the original image.

The predictor uses an adaptive linear filter method based on the work presented by Gersho [9, 11]. Observed quantities x_k, analogous to the quantized prediction residuals of Figure 2.1, which are statistically dependant on the process d_k, analogous to the image samples, produce estimates z_k of the desired samples d_k. A linear preprocessor ϕ takes previously observed quantities x_k and transforms them to a vector $\mathbf{u}_k$ given by

$$\mathbf{u}_k = \Phi(x_k, x_{k-1}, \ldots). \tag{2.1}$$

The elements of $\mathbf{u}_k$ are weighted by a vector $\mathbf{c}_k$ to produce the prediction z_k as

$$z_k = \mathbf{c}_k \cdot \mathbf{u}_k. \tag{2.2}$$

The weight vector is adapted (updated) in increments $\Delta\mathbf{c}_k$ based on observed error residuals $z_k - d_k$ of previous predictions.

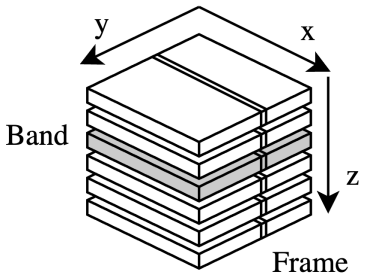


Figure 2.2: Layout of a hyperspectral image. Samples of the same z coordinates belong to the same frequency band, whereas samples of the same y coordinate are denoted as frames. Samples within the same frame of a band are denoted as lines.

Figure 2.2 shows the layout of a hyperspectral image. The sizes of the dimensions x, y, and z are denoted N_x, N_y, and N_z. Samples of the same z coordinate correspond to frequency bands, while samples of the same y coordinate are denoted as part of the same frame. In a bushbroom imager, like the one on-board the HYPSO-1 mission, images are captured one frame at a time [3]. As the satellite moves, subsequent frames are concatenated to form the final image. The camera frame rate controls the time, and thus physical space, between captured frames. Samples within the same frame of a band are referred to as lines, and samples of the same x and y coordinate belong to a pixel.

In order to correctly reproduce the original image, the sample processing order of the compressor and decompressor must be the same. The standard recognizes two main categories of processing orders: Band Sequential (BSQ) and Band Interleaved (BI). BSQ ordering processes samples in coordinate order (x, y, z), meaning all samples of one band are processed before proceeding to the next. BI ordering processes samples frame by frame. The frame interleaving depth M dictates the processing order within the frame. The standard defines two special cases: Band Interleaved by Line (BIL) and Band Interleaved by Pixel (BIP), with interleaving depths equal to 1 and N_z respectively. BIL processes

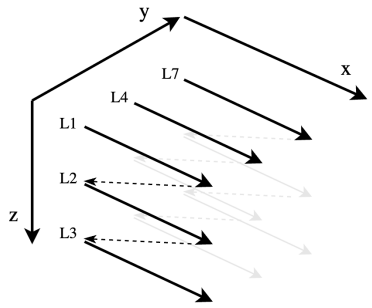


Figure 2.3: BIL processing order. Lines, L1 to L3, within the first frame are processed before proceeding to the next frame.

samples within a frame line by line in coordinate order (x, z, y), as presented by Figure 2.3. BIP processes samples in the order (z, x, y).

Image samples may be referred to by their x, y, and z coordinates as $s(x, y, z)$. To shorten the notation, the standard introduces the spatial index t, given by $t = yN_x + x$. Thus, a sample may be referred to as $s_z(t)$. The number of bits used to represent a sample is the dynamic range D.

The following sections give a brief technical overview of the predictor and encoder stages of the issue 2 algorithm. Symbols are mentioned in text, and are the same as those used in the issue 2 standard unless otherwise is specified. As a quick reference, and for the readers convenience, all symbols are listed in Table 2.1. For a detailed review of the standard please see The Consultative Committee for Space Data Systems, 'Low-Complexity Lossless and Near-Lossless Multispectral and Hyperspectral Image Compression,' CCSDS, Recommended Standard CCSDS 123.0-B-2, 2019.

Table 2.1: Issue 2 compressor symbol table.

Symbol	Name
$s_z(t)$	Image sample
$\sigma_z(t)$	Local sum
$\mathbf{U}_z(t)$	Local difference vector
$d_z(t)$	Central local difference
$\hat{d}_z(t)$	Predicted central local difference
$\check{s}_z(t)$	High-resolution predicted sample
$\tilde{s}_z(t)$	Double-resolution predicted sample
$\hat{s}_z(t)$	Predicted sample
$\Delta_z(t)$	Prediction residual
$m_z(t)$	Maximum error
$q_z(t)$	Signed quantizer index
$s'_z(t)$	Clipped quantizer bin center
$\tilde{s}''_z(t)$	Double-resolution sample representative
$s''_z(t)$	Sample representative
$\theta_z(t)$	Predicted sample and sample endpoint difference
$\delta_z(t)$	Mapped quantizer index
$e_z(t)$	Double-resolution prediction error
$\mathbf{W}_z(t)$	Weight vector
$\tilde{\Sigma}_z(t)$	High-resolution accumulator
$\Gamma(t)$	Counter

2.1.1 Predictor

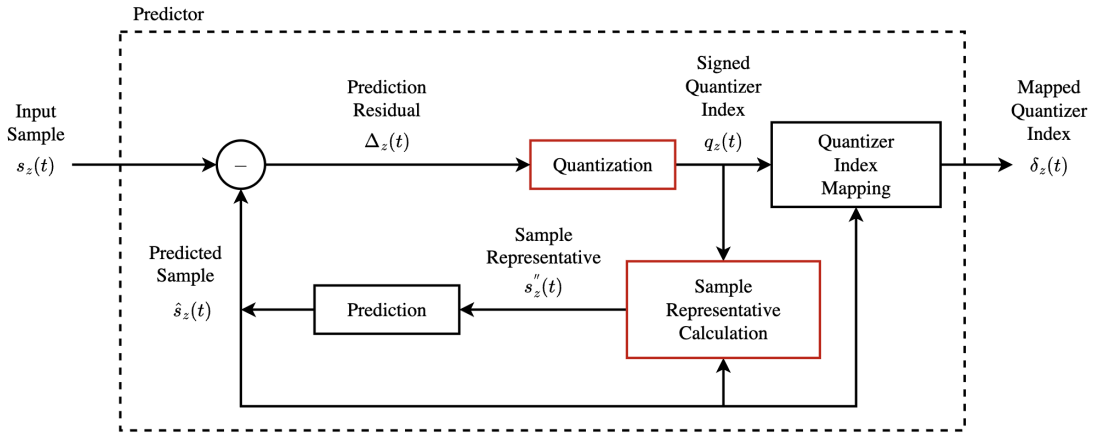


Figure 2.4: Block diagram of the predictor stage of issue 2. The quantization and sample representative steps, marked in red, are new inclusions compared to Issue 1.

Figure 2.4 shows a block diagram of the issue 2 predictor stage with new features as compared to issue 1 marked in red. In the decompressor, the outputs of the predictor stage are first losslessly reproduced by decoding the encoder outputs stored in the compressed bitstream. In order to correctly retrieve the original image, the decompressor must be able to reproduce the prediction model of the compressor. Normally, this would require the samples from the original image to be known. To solve this, the issue 2 predictor makes use of sample representatives $s''_z(t)$ which may be calculated directly from the quantized prediction residuals, and thus available when decompressing.

Quantization

Prediction residuals $\Delta_z(t)$ are quantized by a uniform quantizer producing quantizer indices $q_z(t)$ given by

$$q_z(t) = \begin{cases} \Delta_z(0), & t = 0, \\ \operatorname{sgn}(\Delta_z(t)) \left\lfloor \dfrac{|\Delta_z(t)| + m_z(t)}{2m_z(t)+1} \right\rfloor, & t > 0, \end{cases} \tag{2.3}$$

where the prediction residual is the difference between input samples $s_z(t)$ and predicted sample values $\hat{s}_z(t)$ given by

$$\Delta_z(t) = s_z(t) - \hat{s}_z(t). \tag{2.4}$$

Figure 2.5 shows a uniform quantizer centered at zero. Inputs are mapped to bins with size Δ, also called the step size [12]. The step size of the issue 2 quantizer is $2m_z(t)+1$ and is controlled by the maximum error value $m_z(t)$. This guaranties that the prediction residual can be reconstructed with an error of at most $\pm m_z(t)$. A quantizer index of zero means that the predicted sample was within $m_z(t)$ of the actual sample. Positive and negative quantizer indices indicate that the prediction overshot or undershot the sample, respectively. The clipped quantizer bin center $s'_z(t)$ is used for calculation of sample representatives, and is given by

$$s'_z(t) = \operatorname{clip}\big(\hat{s}_z(t) + q_z(t)\big(2m_z(t)+1\big),\ \{s_{\min}, s_{\max}\}\big), \tag{2.5}$$

where s_{min} and s_{max} are the lower and upper bounds for the input sample value dictated by the dynamic range D.

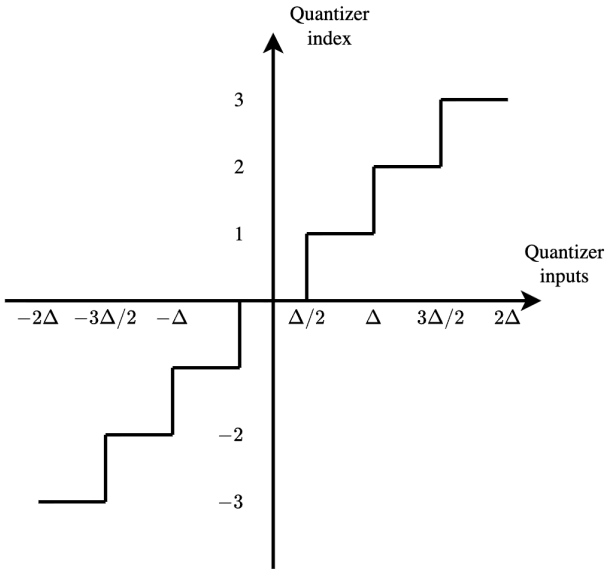


Figure 2.5: Uniform quantizer with step size Δ.

The user may select an absolute and a relative error limit, a_z and r_z respectively, giving fine tuned control over the per pixel difference as compared to the original image. To losslessly encode image samples, the maximum error limit may be set to zero. When error limits are used, the maximum error is given by

$$m_z(t) = \min\left(a_z, \left\lfloor \frac{|r_z \hat{s}_z(t)|}{2^D} \right\rfloor\right). \tag{2.6}$$

TODO: periodic error limit updating.

Because the codewords used by the entropy coder to produce the compressed bitstream are only defined for positive integers, the signed quantizer indices are mapped to positive integer mapped quantizer indices $\delta_z(t)$. This mapping is reversible and is given by

$$\delta_z(t) = \begin{cases} |q_z(t)| + \theta_z(t), & |q_z(t)| > \theta_z(t), \\ 2|q_z(t)|, & 0 \le (-1)^{\tilde{s}_z(t)} q_z(t) \le \theta_z(t), \\ 2|q_z(t)| - 1, & \text{otherwise}, \end{cases} \tag{2.7}$$

where the double-resolution predicted sample value $\tilde{s}_z(t)$ is an intermediate value produced by the prediction stage (Figure 2.4) which will be covered later in the document. The predicted sample and sample endpoint difference $\theta_z(t)$ is given by

$$\theta_z(t) = \begin{cases} \min\{\hat{s}_z(0) - s_{\min},\ s_{\max} - \hat{s}_z(0)\}, & t = 0, \\ \min\left(\left\lfloor \frac{|\hat{s}_z(t) - s_{\min} + m_z(t)|}{2m_z(t)+1} \right\rfloor, \left\lfloor \frac{|s_{\max} - \hat{s}_z(t) + m_z(t)|}{2m_z(t)+1} \right\rfloor\right), & t > 0. \end{cases} \tag{2.8}$$

Sample representatives

As mentioned, sample representatives are used in place of the sample values in order to allow the decompressor to reproduce the prediction model. Selecting sample representatives to be equal to the quantizer bin center minimizes the per-pixel reconstruction error. However, a different selection may be used to minimize other quality metrics, such as the Mean Square Error (MSE) (section 2.3) [9]. For this reason, the user is provided with a set of parameters which control the selection of the

sample representatives in relation to the quantizer bin center. Intermediary values of the sample representative calculation are the double-resolution sample representative $\tilde{s}''_z(t)$ and the quantizer bin center $s'_z(t)$.

Local Sums and Differences

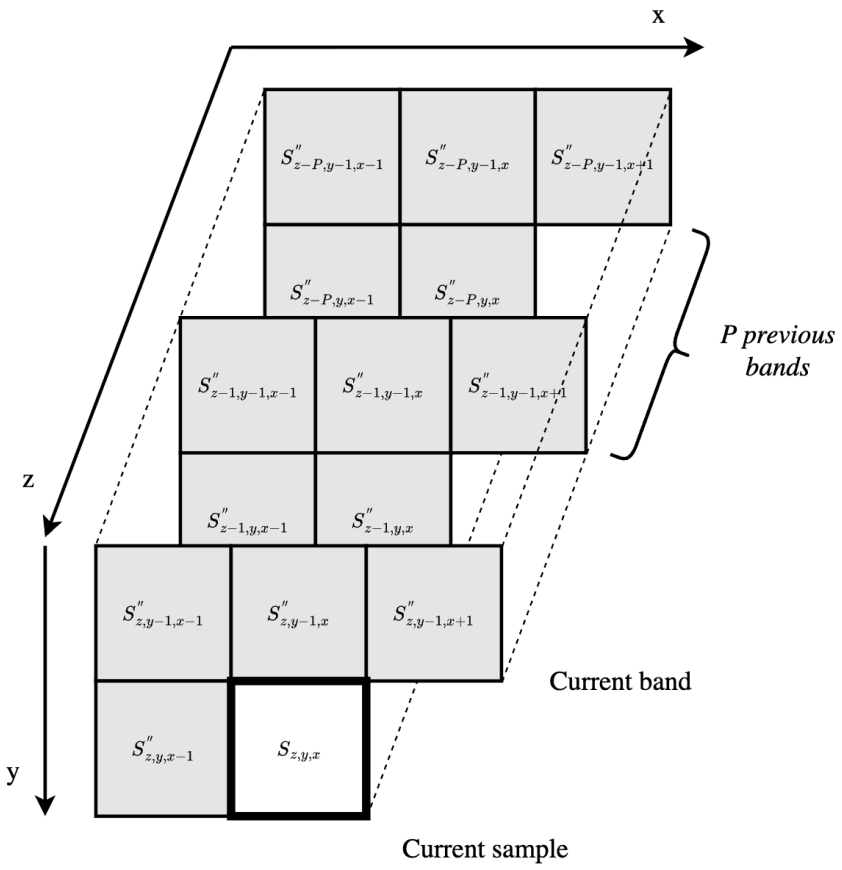


Figure 2.6: A typical prediction neighborhood. Local sums are used to find local differences in P previous bands (prediction bands) which are used to find the predicted sample.

Sample representatives of previously processed samples form the prediction neighborhood as shown in Figure 2.6. The parameter P controls the number of spectral bands that are included in the prediction calculation, called prediction bands. A weighted sum of spatially adjacent samples to the current sample form the local sum $\sigma_z(t)$. Issue 2 defines two categories of local sum configurations: column-oriented and neighbor-oriented, both of which can be configured as wide and narrow. Figure 2.7 shows the wide and narrow configurations for neighbor-oriented local sums. Full details and corner cases of the local sum configurations are not important for this discussion, but as an example, narrow neighbor-oriented local sums are defined as

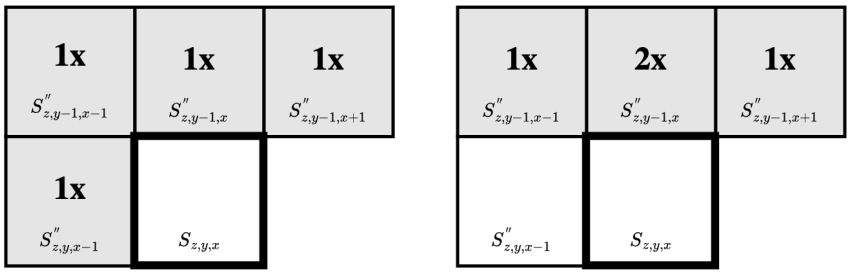


Figure 2.7: Wide and narrow neighbor-oriented local sum configurations.

$$\sigma_{z,y,x} = \begin{cases} s''_{z,\,y-1,\,x-1} + 2s''_{z,\,y-1,\,x} + s''_{z,\,y-1,\,x+1}, & y > 0,\ 0 < x < N_X - 1, \\ 4s''_{z-1,\,y,\,x-1}, & y = 0,\ x > 0,\ z > 0, \\ 2\Big(s''_{z,\,y-1,\,x} + s''_{z,\,y-1,\,x+1}\Big), & y > 0,\ x = 0, \\ 2\Big(s''_{z,\,y-1,\,x-1} + s''_{z,\,y-1,\,x}\Big), & y > 0,\ x = N_X - 1, \\ 4s_{\text{mid}}, & y = 0,\ x = 0,\ z = 0. \end{cases} \tag{2.9}$$

The (central) local difference $d_z(t)$ is the difference between the local sum and the current sample. Local differences of the prediction bands form the local difference vector $\mathbf{U}_z(t)$, and is analogous to the vector $\mathbf{u}_k$ (Equation 2.1) of the adaptive linear filter. The vector is defined as

$$\mathbf{U}_z(t) = \begin{bmatrix} d_{z-1}(t) \\ d_{z-2}(t) \\ \vdots \\ d_{z-P^*_z}(t) \end{bmatrix}, \tag{2.10}$$

where P^*_z is the number of used prediction bands found by

$$P^*_z = \min(z, P). \tag{2.11}$$

Two different prediction modes are defined in the standard: full and reduced. The above definition of the local difference vector is that of the reduced mode. During full prediction mode, three additional local difference values $d^N_z(t)$, $d^W_z(t)$ and $d^{NW}_z(t)$ are added to the vector, named directional local differences. These are calculated as the weighted difference between sample representatives in three directions and the local sum, and add avoidable dependencies on neighbouring sample representatives. For the full definition see the standard.

Weight Updates

The predicted central local difference $\hat{d}_z(t)$ is computed as

$$\hat{d}_z(t) = \mathbf{W}_z(t) \cdot \mathbf{U}_z(t). \tag{2.12}$$

Here $\mathbf{W}_z(t)$ is the weight vector, which for reduced prediction mode is defined as

$$\mathbf{W}_z(t) = \begin{bmatrix} \omega^{(1)}_z(t) \\ \omega^{(2)}_z(t) \\ \vdots \\ \omega^{(P^*_z)}_z(t) \end{bmatrix}, \tag{2.13}$$

where $\omega_z^{(i)}(t)$ is the weight corresponding to the ith prediction band. As with the local difference vector, three additional weights are used during full prediction mode.

Weights may be initialized using two methods: default and custom initiaization. Custom initialization allows the user to the provide per-band initial weight values. This may result in better compression performance when the image sensor characteristics or other image metrics are know. For default initiaization the weights are initialized as

$$\omega_z^{(1)}(1) = \frac{7}{8}2^{\Omega}, \qquad \omega_z^{(i)}(1) = \left\lfloor \frac{1}{8}\,\omega_z^{(i-1)}(1) \right\rfloor, \quad i = 2, 3, \ldots, P_z^*. \tag{2.14}$$

Weights are represented as signed integers of $\Omega+3$ bits, where Ω is the weight bit resolution. This gives a range of possibles weight values between ω_{min} and ω_{max} as

$$\omega_{\min} = -2^{\Omega+2},\ \omega_{\max} = -2^{\Omega+2} - 1. \tag{2.15}$$

Weights are updated for every sample based on the performance of the prediction model. Using the notation presented by Jia et al. in [4] we define the weight offset as

$$\omega_{z,\text{offset}}^{(i)}(t) = \left\lfloor \frac{1}{2}\left(2^{-\rho(t)} \cdot d_{z-i}(t) \cdot \text{sgn}^{+}[e_z(t)] + 1\right) \right\rfloor, \tag{2.16}$$

where the double-resolution prediction error $e_z(t)$ is found using the quantizer bin center and double-resolution predicted sample value as

$$e_z(t) = 2s_z'(t) - \tilde{s}_z(t). \tag{2.17}$$

The user can control how quickly the prediction model adapts to rapid changes in the sample values by use of the weight update scaling exponent, given by

$$\rho(t) = \text{clip}\left(\nu_{\min} + \left\lfloor \frac{t - N_x}{t_{\text{inc}}} \right\rfloor, \{\nu_{\min}, \nu_{\max}\} \right) + D - \Omega, \tag{2.18}$$

where ν_{min}, ν_{max}, and t_{inc} are user tunable parameters. The final updated weight is calculated as

$$\omega_z^{(i)}(t+1) = \text{clip}\left(\omega_z^{(i)}(t) + \omega_{z,\text{offset}}^{(i)}(t), \{\omega_{min}, \omega_{max}\} \right). \tag{2.19}$$

Prediction Calculation

The predicted central local difference $\hat{d}_z(t)$ in combination with the local sum $\sigma_z(t)$ produces the high-resolution predicted sample value $\breve{s}_z(t)$ as

$$\breve{s}_z(t) = \text{clip}\left(\begin{array}{l} \text{mod}_R^*\left[\hat{d}_z(t) + 2^{\Omega}\left(\sigma_z(t) - 4s_{\text{mid}}\right)\right] \\ \quad + 2^{\Omega+2}s_{\text{mid}} + 2^{\Omega+1},\ \left\{2^{\Omega+2}s_{\min},\ 2^{\Omega+2}s_{\max} + 2^{\Omega+1}\right\} \end{array} \right). \tag{2.20}$$

where R is a user tunable parameter and s_{mid} is the middle sample value, dictated by the dynamic range D. From this, the double-resolution predicted sample value $\tilde{s}_z(t)$ is obtained as

$$\tilde{s}_z(t) = \begin{cases} \left\lfloor \dfrac{\breve{s}_z(t)}{2^{\Omega+1}} \right\rfloor, & t > 0, \\ 2s_{z-1}(t), & t = 0,\ P > 0,\ z > 0, \\ 2s_{\text{mid}}, & t = 0 \text{ and } (P = 0 \text{ or } z = 0). \end{cases} \tag{2.21}$$

Finally, the predicted sample value $\hat{s}_z(t)$ is given by

$$\hat{s}_z(t) = \left\lfloor \frac{\tilde{s}_z(t)}{2} \right\rfloor. \tag{2.22}$$

Decompression

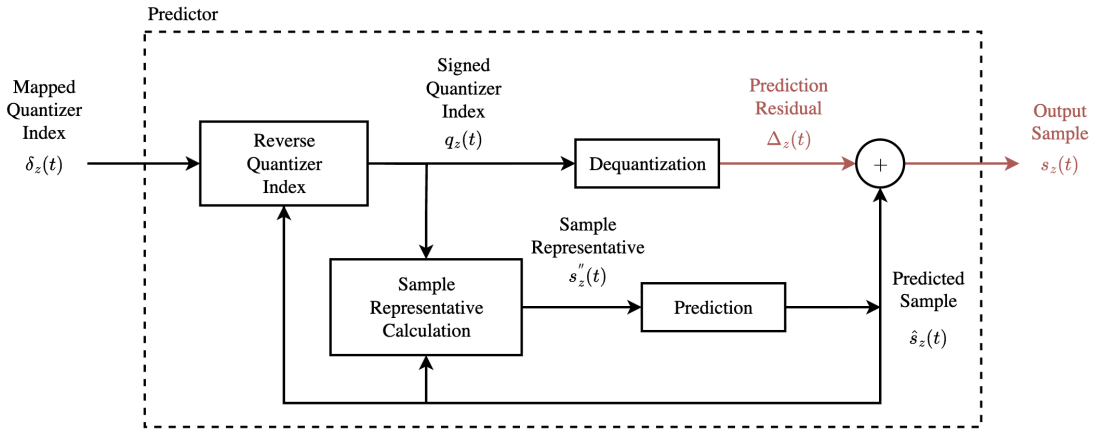


Figure 2.8: Block diagram of the predictor stage of the decompressor. The prediction model is identical to that of the compressor, except for the prediction residual and reconstructed sample during near-lossless mode, marked in red.

Figure 2.8 shows a block diagram of the predictor stage of the decompressor, which replicates the prediction model of the compressor. Quantizer indices are reconstructed without loss from the mapped quantizer indices as

$$q_z(t) = \begin{cases} (\theta_z(t) - \delta_z(t))\ \mathrm{sgn}^+\big(\hat{s}_z(t) - s_{\mathrm{mid}}\big), & \delta_z(t) > 2\theta_z(t), \\ \left\lfloor \dfrac{\delta_z(t)+1}{2} \right\rfloor (-1)^{\tilde{s}_z(t)+\delta_z(t)}, & \delta_z(t) \le 2\theta_z(t), \end{cases} \tag{2.23}$$

where $\hat{s}_z(t)$ and $\theta_z(t)$ are computed using prediction data from previously decoded samples. The range of possible values for the reconstructed sample is now $[s'_z(t) - m_z(t), s'_z(t) + m_z(t)]$ [9]. When the maximum error $m_z(t)$ is zero, prediction residuals may be reconstructed without loss. For this work the quantizer bin center $s'_z(t)$ is selected for reconstruction of samples. Thus, the dequantization step is given by

$$\Delta_z(t) = \begin{cases} q_z(0), & t = 0, \\ \mathrm{sgn}(q_z(t))|q_z(t)|(2m_z(t)+1), & t > 0. \end{cases} \tag{2.24}$$

Finally, the reconstructed sample is retrieved from the predicted sample value and dequantized prediction residual as

$$s_z(t) = \mathrm{clip}(\Delta_z(t) + \hat{s}_z(t),\ \{s_{\min}, s_{\max}\})\,. \tag{2.25}$$

Data Dependencies

Both the issue 1 and issue 2 algorithms contain backwards data dependencies that make efficient implementations for hardware challenging. Dependencies are split into two groups: spectral and spatial dependencies, referred to as $z-1$ and $t-1$ dependencies. As an example take the calculation of the weight vector, which is present in both versions of the algorithm, and for issue 2 is given by equations 2.16 and 2.19. This involves a $t-1$ dependency on the weight vector of sample $s_z(t-1)$. Pipelined hardware implementation must wait for such dependencies to resolve before proceeding, placing a restriction on the maximum achievable throughput.

The selection of sample processing order affects the number of processed samples that interleaves data dependant calculations. Thus, careful selection of processing order may alleviate the problem of data dependencies for pipelined hardware implementations. For the three main processing orders, the number of interleaving samples for $z-1$ and $t-1$ dependencies is summarized in Table 2.2, and may be calculated as follows:

1. BSQ iterates samples in the order (x, y, z). This means that $N_x \cdot N_y$ samples interleave $z-1$ dependencies as all samples of the first band are processed before moving to the next band. Spatial samples are processed sequentially, leaving a single sample between $t-1$ dependencies.
2. BIP iterates samples in the order (z, x, y), leaving a single sample between $z-1$ dependencies. All samples of the same pixel are processed before moving the next spatial index, giving N_z samples between $t-1$ dependencies.
3. BIL iterates samples in the order (x, z, y). Lines are processed sequentially, leaving N_x interleaved samples for $z-1$ dependencies and a single for $t-1$.

Table 2.2: Number of samples interleaving $z-1$ and $t-1$ dependencies for the three main processing orders.

	BSQ	**BIP**	**BIL**
$z-1$	$N_x \cdot N_y$	1	N_x
$t-1$	1	N_z	1

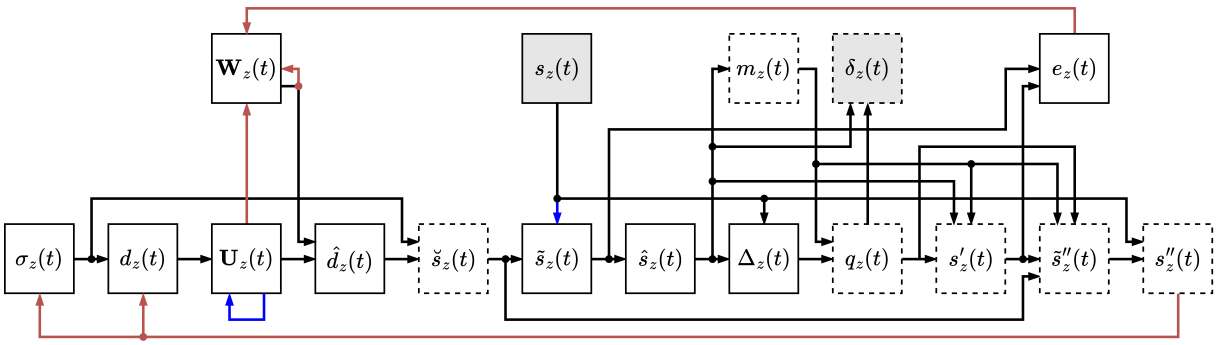


Figure 2.9: Data dependencies of the issue 2 predictor with $t-1$ and $z-1$ dependencies marked in red and blue respectively. Novel stages as compared to issue 1 are marked with stippled lines.

Figure 2.9 gives an overview of the data dependencies present in the issue 2 predictor, with $z-1$ and $t-1$ dependencies marked in blue and red respectively. Novel features compared to issue 1, mainly revolving around the quantization and sample representatives, are marked with stippled lines. In addition to the aforementioned dependency, weight updates form $t-1$ dependencies on the local difference vector $\mathbf{U}_z(t)$ and the double-resolution prediction error $e_z(t)$ (Equation 2.16). The latter depends on results from both the prediction calculation and sample representative calculation (Equation 2.17), presenting a particularly long path for the dependency to resolve compared to the shorter path of issue 1.

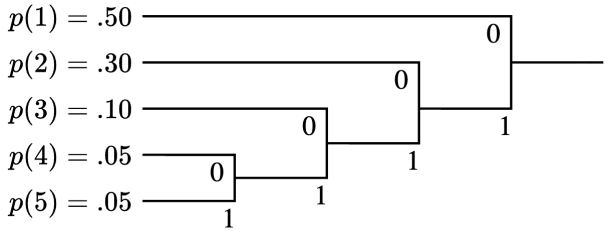
For the wide local sum configurations (Figure 2.7), calculation of local sums $\sigma_z(t)$ form a $t-1$ dependency on sample representatives $s''_z(t)$ (Equation 2.9). Furthermore, directional local differences have a $t-1$ dependence on sample representatives during full prediction mode. The local difference vector $\mathbf{U}_z(t)$ depends on local differences of the prediction bands, forming a $z-1$ dependency. Finally, the double-resolution predicted sample value has a $z-1$ dependency on the sample value from the

previous band (Equation 2.21).

2.1.2 Entropy Coder

The mapped quantizer indices produced by the predictor are passed to an entropy coder where they are losslessly encoded and appended to the compressed bitstream. Entropy coders use sample statistics combined with predefined, variable-length codewords to encode samples with high probability with fewer bits than low probability samples. In this way, samples may be encoded using fewer bits than would be required if encoding the samples at face value.

A commonly used form of entropy encoding is *Huffman coding*. When the probabilities p of a finite set of messages are known, the method composes a set of codewords with a minimized average length [12]. A binary tree is constructed by recursively joining messages of lowest probability, as shown on the left side of Figure 2.10, producing the codewords of the accompanying table. As an example, the string of messages (denoted by index) $\{2, 3, 1, 1, 2, 5\}$ may be encoded using the string of codewords $\{10, 110, 0, 0, 10, 1111\}$, giving the bitstream 1011000101111. Codewords are prefix-free, meaning that no codeword is the prefix of another. This way, an encoded bitstream can be uniquely decoded by traversing from the start of the stream.



	Codeword
$p(1) = .50$	0
$p(2) = .30$	10
$p(3) = .10$	110
$p(4) = .05$	1110
$p(5) = .05$	1111

Figure 2.10: Huffman encoding tree and associated codeword table. Decoding of the prefix free codes may be performed by traversing from the top of the tree until a match is found at one of its branches. Figure recreated from [12].

Issue 1 supports two different entropy coders: the sample-adaptive and block-adaptive encoders. The former uses variable-length codewords, whereas the latter uses a different technique which will not be covered in this work. The minimum length of a variable-length codeword is one bit, meaning that encoding bitrates of less than one bit per sample are unachievable using the sample-adaptive encoder. Due to the low entropy of mapped quantizer indices when using the issue 2 near-lossless mode, a new entropy coder was introduced. This encoder, the *Hybrid encoder*, is able to achieve sub-one bitrates using a combination of the techniques of the sample-adaptive encoder and Variable-to-Variable (V2V) codes. The following overview of the hybrid encoder is based on the work presented by Blanes et al., which gives great insight into the development of and inner workings of the encoder [13].

Hybrid Encoder

Similarly to the sample-adaptive encoder, the hybrid encoder uses adaptive statistics to select between encoding techniques. By adding processed samples to an accumulator while keeping count of the number of processed samples, an estimate of the average sample value may be obtained by dividing the accumulator by the counter. At certain intervals the accumulator and counter are divided by two. This reduces the impact of previously processed samples while preserving the average estimate. Users may change the length of this interval by use of the rescaling counter size γ^* parameter.

The high-resolution accumulator $\tilde{\Sigma}_z(t)$ keeps a separate accumulator for each band, whereas the counter $\Gamma(t)$ is shared between bands. They are given by

$$\tilde{\Sigma}_z(t) = \begin{cases} \tilde{\Sigma}_z(t-1) + 4\delta_z(t), & \Gamma(t-1) < 2^{\gamma^*} - 1, \\ \left\lfloor \dfrac{\tilde{\Sigma}_z(t-1) + 4\delta_z(t) + 1}{2} \right\rfloor, & \Gamma(t-1) = 2^{\gamma^*} - 1, \end{cases} \tag{2.26}$$

and

$$\Gamma(t) = \begin{cases} \Gamma(t-1) + 1, & \Gamma(t-1) < 2^{\gamma^*} - 1, \\ \left\lfloor \dfrac{\Gamma(t-1) + 1}{2} \right\rfloor, & \Gamma(t-1) = 2^{\gamma^*} - 1. \end{cases} \tag{2.27}$$

Based on the average estimate $\tilde{\Sigma}_z(t)/\Gamma(t)$, the hybrid encoder chooses between two different types of encodings: high-entropy and low-entropy codes. For $t = 0$, the mapped quantizer index is encoded as is using an unsigned integer of D bits. For $t > 0$, if $\tilde{\Sigma}_z(t) \cdot 2^{14} \geq T_0 \cdot \Gamma(t)$, a high-entropy code is selected. Otherwise, a low-entropy code is used. Here, T_0 is the low-entropy threshold.

Low-entropy codes use a set of 16 V2V codes, indexed by the code index i. The work by Blanes et al. shows that the statistical distribution of low-entropy mapped quantizer indices produced during near-lossless mode deviate from the geometric distribution of high-entropy indices. To accommodate this, a set of V2V where developed, allowing efficient encoding of the low-entropy codes. Each of the 16 codes has a corresponding codeword table, similar to the Huffman code example of Figure 2.10.

By use of the average estimate, and a threshold T_i, the low-entropy code index i is selected. When multiple indices of the same index are observed, possibly interleaved by high-entropy codes, they are accumulated (active prefix) until a matching codeword is found and appended to the bitstream. This allows multiple mapped quantizer indices to be encoded by a single codeword. Low-entropy codewords corresponding to multiple samples may appear in the bitstream several samples after the first sample it encodes. For this reason, decoding can no longer proceed in the forward direction. To allow backwards decoding, the V2V codes are suffix-free, rather than prefix-free. This means that no codeword is the suffix of another.

High-entropy indices are encoded using a set of 29 Golomb power-of-two (GPO2) codes. First described by Golomb in [14], Golomb codes device a method for producing optimal prefix-free codewords for a set of messages following a geometric distribution. By using parameters which are powers of two, thereby Golomb *power-of-two*, operations required by hardware implementations are simplified at a slight loss in encoding performance. Additionally, the maximum codeword length is limited to $U_{max} + D$ bits, where U_{max} is the unary length limit, to further simplify hardware and cost of encoding outliers. To facilitate the backwards decoding required by low-entropy codes, the GPO2 are reversed to produce suffix-free codes.

In order to retrieve the mapped quantizer indices from the compressed bitstream, the decoder must be able to reproduce the average estimates of the encoder. For this reason, a compressed image tail encodes the final accumulator values and state of the V2V active prefixes. During encoding, the average estimate is updated before encoding the sample. This allows the decoder to use the current average estimate to select high- or low-entropy decoding before updating the estimate.

2.1.3 Header and Optional Files

Optional tables, error limit table, encoder accumulator initial file, and error limits encoded in bitstream.

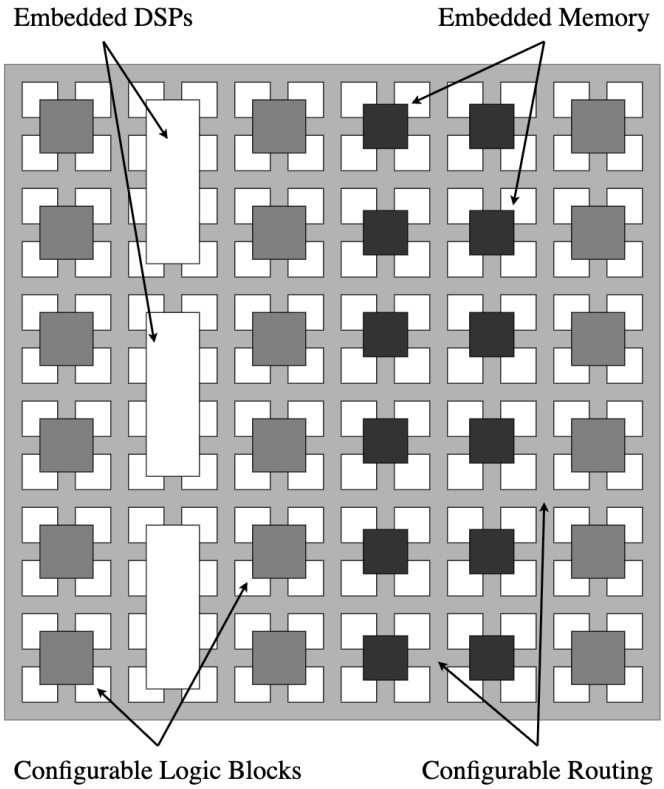


Figure 2.11: General layout of an FPGA. CLBs, DSPs, and memory blocks are placed in a configurable routing network.

2.2 Hardware Accelerators [1]

Embedded real-time systems often operate under strict deadlines. Missing these deadlines can be detrimental to system performance and correct functioning. At the same time, designers must balance budgets on power, area and cost. An effective approach to designing robust and responsive systems is to utilize multiprocessors [15]. Multiprocessor systems contain multiple Processing Elements (PE), that can be optimized for specific functions, performing tasks in parallel. It is well known that the use of simpler units running in parallel can provide the same performance at a lower power and cost [15, 16].

In addition to programmable PEs, e.g. Central Processing Units (CPU), which provide great flexibility, multiprocessor systems can achieve enormous performance improvements by the use of specialized hardware accelerators [15]. One such example is the Graphical Processing Unit (GPU), which has become a staple of modern personal computers. FPGAs allow designers to design custom logic at low costs, as compared to the higher performance and costly Application Specific Integrated Circuits (ASIC). Most FPGAs also come at the added benefit of being reconfigurable, which provides flexibility for remotely deployed systems.

Figure 2.11 shows the layout of a typical FPGA. By use of Look-Up-Tables (LUT) and Flip-Flops (FF), Configurable Logic Blocks (CLB) can be configured to provide any logic function. Units are mounted in a large routing matrix, allowing custom routing between the CLBs. These two mechanisms provide the core functionality of the FPGA, and allow the implementation of any logic design. In order to increase efficiency in area, power and performance, it is common to add additional, specialized hardware units. Digital Signal Processors (DSP) provide common arithmetic functions, while embedded Block

[1]The section on hardware accelerators is taken as is from project thesis (TODO: make proper cite)

Random Access Memory (BRAM) provides efficient data storage. The hardware resource utilization of a design for FPGA is typically measured in terms of the amount of LUTs, FFs, DSPs, and BRAM used.

Another way of utilizing parallelization to increase performance is pipelining. By separating data independent stages of the calculation with registers, known as pipeline registers, the clock frequency can be increased while still issuing new data for processing every cycle. This approach is widely used in hardware designs, for example in CPUs. The clock frequency is limited by the longest path between any two pipeline registers. For implementations of the issue 2 algorithm, the aforementioned data dependencies create problems when trying to achieve high degrees of pipelining.

Compared to software design, hardware design comes at increased complexity in terms of design and development. A typical design starts at a microarchitectural level before the design is converted to the Register Transfer Level (RTL) using languages like VHDL, Verilog and Systemverilog. RTL code is then synthesized to a gatelevel netlist which, through a process called Place and Route (PnR), is mapped to a specific FPGA or for ASIC. To ensure correct functioning of the final result, extensive verification must be performed at all steps in the design process.

To alleviate the challenges in long design time of RTL code, a new design flow has emerged named High-Level Synthesis (HLS). The advantage of using the HLS approach is reduced development time, as the algorithm can be described in a high-level language like C or C++, which is then converted to RTL [17]. However, the tools for such a conversion create suboptimal results when compared to writing the implementation directly in RTL using languages like VHDL or Systemverilog, giving the designer fine-grained control over optimizations.

The satellites of the HYPSO project use Multiprocessor System-on-Chips (MPSoC) from AMD (formerly Xilinx), namely the Zynq chips [3]. These COTS embed two ARM processing units, a simple GPU, and several useful peripherals including Analog-to-Digital Converters (ADC), timers and support for various communication protocols. The system runs the Linux operating system, providing tools for management of complex tasks and threads. Additionally, the chips include embedded Programmable Logic (PL), equivalent to an embedded FPGA. Embedding the PL directly in the MPSoC allows custom logic to directly access the main data buss and other systems on the chip, including Direct Memory Access (DMA) for fast and CPU independent memory accesses. These features give great benefits in terms of efficiency in speed and power as compared to separate FPGAs.

TODO: paragraph about verification. DUT, constrained random, golden model etc

TODO: paragraph about bus interfaces

2.3 Coding Performance and Decompressed Image Quality

$$\mathrm{CR} = \frac{N_x \cdot N_y \cdot N_z \cdot D}{\text{compressed data size (bits)}}. \tag{2.28}$$

$$\mathrm{PSNR}_z = 10 \cdot \log_{10}\left(\frac{\mathrm{MAX}_I^2}{\mathrm{MSE}_z}\right), \tag{2.29}$$

$$\mathrm{MSE}_z = \frac{1}{N_x N_y} \sum_{i=0}^{N_x-1} \sum_{j=0}^{N_y-1} \left[I_z(i,j) - K_z(i,j)\right]^2. \tag{2.30}$$

$$\mathrm{PSNR}_{HSI} = \frac{1}{N_z} \sum_{z=0}^{N_z-1} \mathrm{PSNR}_z. \tag{2.31}$$

2.4 Software Optimization

Compiled vs interpreted languages. Amdals law

Chapter 3

State-of-the-Art

Table 3.1: The number of processed samples interleaving $z-1$ and $t-1$ dependencies, including the novel FID ordering in addition to the standard orderings.

	BSQ	**BIP**	**BIL**	**FID**
$z-1$	$N_x \cdot N_y$	1	N_x	N_z-1
$t-1$	1	N_z	1	N_z

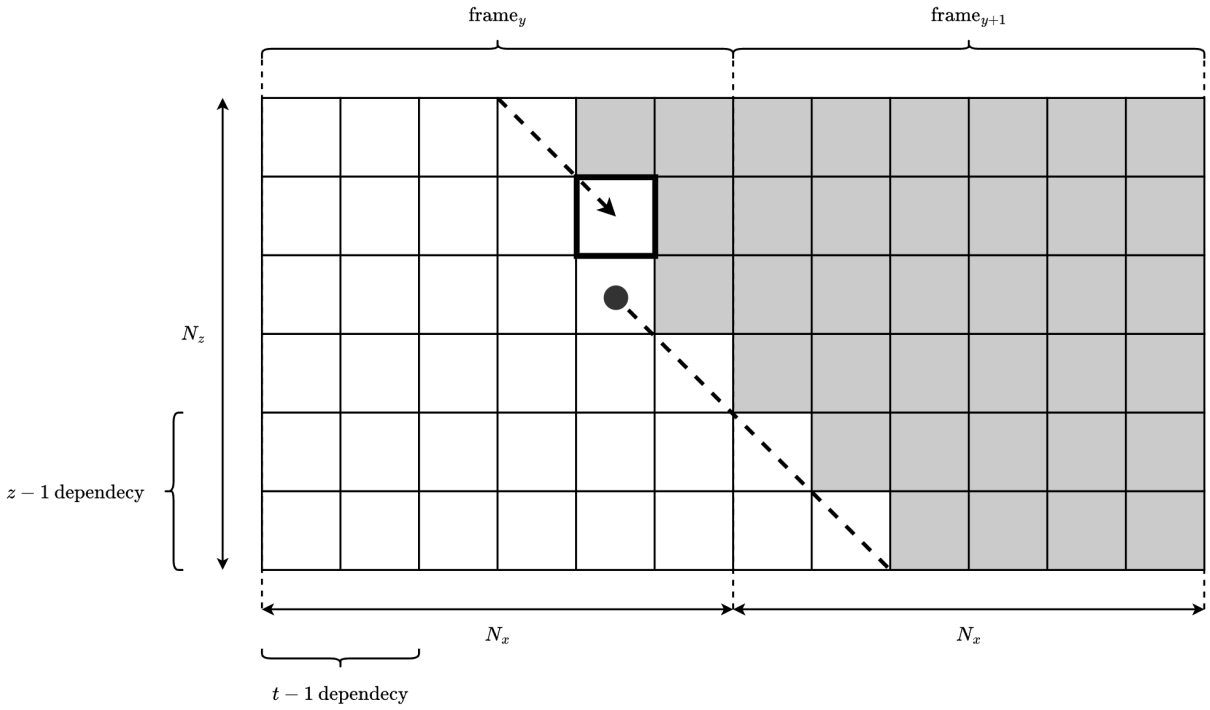


Figure 3.1

$$\operatorname{sgn}^{+}\left[e_z(t)\right] = \begin{cases} -1, & \text{if } |s_z(t)-\hat{s}_z(t)| \leq m_z(t) \text{ and } \tilde{s}_z(t) \text{ is odd,} \\ 1, & \text{if } |s_z(t)-\hat{s}_z(t)| \leq m_z(t) \text{ and } \tilde{s}_z(t) \text{ is even,} \\ \operatorname{sgn}^{+}\left[s_z(t)-\hat{s}_z(t)\right], & \text{otherwise.} \end{cases} \tag{3.1}$$

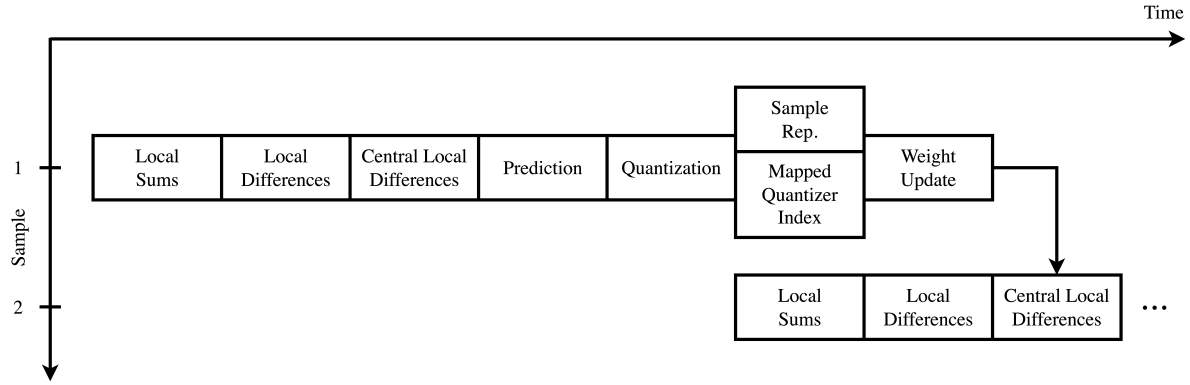


Figure 3.2

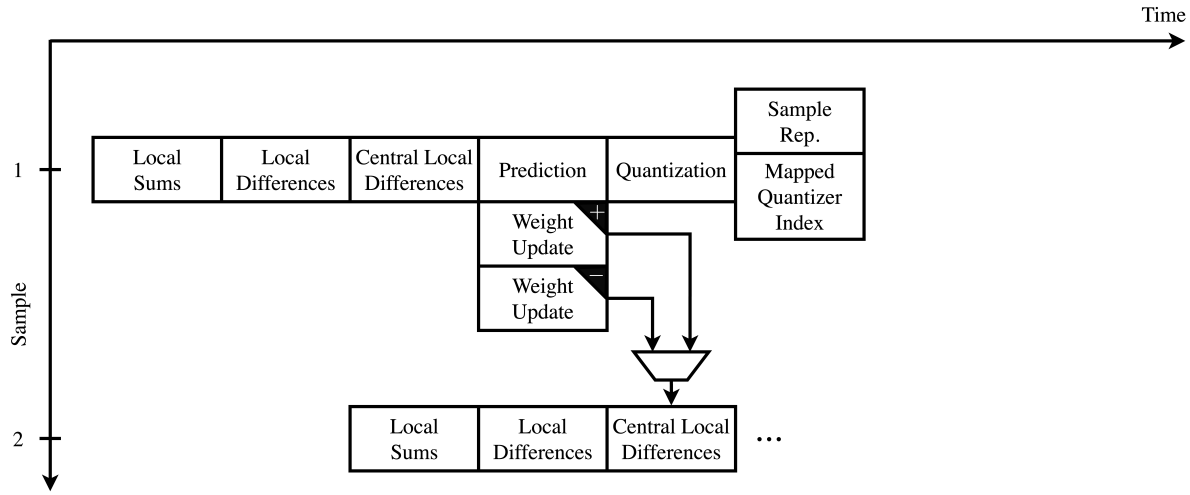


Figure 3.3

$$\omega^{(i)}_{z,\text{unclipped}}(t+1) = \begin{cases} \omega^{(i)}_z(t), & x \leq 2, \\ \omega^{(i)}_z(t-3) + \omega^{(i)}_{z,\text{offset}}(t-3), & x = 3, \\ \left\lfloor \frac{1}{2}\left(\omega^{(i)}_z(t-3) + \omega^{(i)}_{z,\text{offset}}(t-3) + \omega^{(i)}_z(t)\right) \right\rfloor, & \text{otherwise.} \end{cases} \tag{3.2}$$

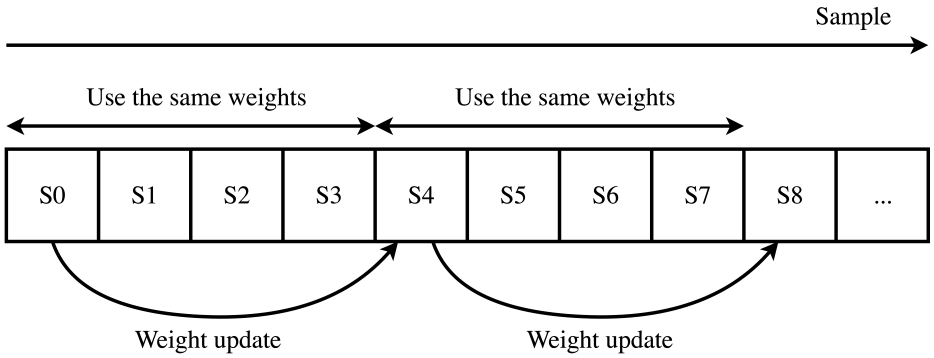


Figure 3.4: The proposed weight update scheme by Jia et al. Weights are kept unchanged for the first samples of every line, after which the sample from four samples earlier is used in the weight update calculation.

Table 3.2: Throughputs of hardware implementations of the issue 2 algorithm in open literature.

Implementation	**FPGA**	**Throughput**
Barrios et al. [17]	XCKU040	18 MS/s
Sánchez et al. [18, 19], predictor	XCKU040	231 MS/s
Barrios et al. [19], predictor	XCKU040	236 MS/s
Báscones et al. [20]	XQRKU060	285 MS/s
Chatziantoniou et al. [21] 1 core	XCKU040	285 MS/s
Chatziantoniou et al. [21] 5 core	XCKU040	1375 MS/s
Vorhaug et al. [5]	XCKU040	110 MS/s
Jia et al. [4], predictor	XCZU5EV	348 MS/s

Chapter 4

Previous Work

This work builds upon, and aims to improve, the work of Vorhaug in [5]. Vorhaug implemented an issue 2 compliant hardware accelerator for use on-board the HYPSO missions. The hardware accelerator was successfully integrated and tested on-board the HYPSO-1 satellite, showing its viability for use. Before testing, the hardware was verified using a verification model, also written by Vorhaug. The following sections outlines the verification model and design of the hardware accelerator.

4.1 Verification Model

The verification model is implemented in the Python programming language and is fully compliant with the issue 2 standard. This includes support for all three entropy coders, full and reduced prediction mode, and all available local sum types. Since the purpose of the model is hardware verification, it saves intermediate values of the calculations and stores them in files. This results in large amounts of data to be stored in memory during execution, as well as written to the disk after completion. For this reason, the model is not suited for compression of full scale satellite images. Note that the verification model only implements compression, not decompression.

The model uses an object oriented programming style and consists of, in addition to a command line interface, three main modules: the header reader, predictor, and encoder. The header class stores the configuration used by the algorithm. Additionally it supports reading the configuration from files containing a header binary, optional tables and error limits. The raw image input of the compressor is passed to the predictor class which produces the mapped quantizer indices. These are passed to one of three encoder classes, depending on the selected entropy encoder. The header binary, produced by the header class, is prepended to the compressed image body before the result is stored to a file.

As the model is only intended for small test images, the calculations are implemented in a fully serial manner. All intermediary results are stored in global arrays that are modified in place. This removes the need for careful buffering of data that is needed by later calculations at the cost of massive penalties in memory usage. The use of large, global arrays also impacts execution time due to the constantly moving large chunks of data in and out the the memory caches. A final caveat is that modifying and accessing data in place makes the flow of the program harder to read, as function calls do not have clearly defined inputs and outputs at the call site.

4.2 Hardware Accelerator

The hardware accelerator uses the mathematical optimization approach suggested by Sánchez et al. Figure 4.1 shows a block diagram of the implementation. The implementation is only configurable using parameters in the RTL code, and not at runtime. This allows high optimization by the synthesizer and PnR tool, resulting in a small hardware area. Data is moved in and out of the module using

the AXI bus interface, which allows easy integration into the MPSoC on-board HYPSO. Because the implementation uses the BIL processing order, as required by the mathematical optimization approach, the design features an optional reordering buffer. This module reorders from the BIP used by the systems on HYPSO to BIL ordering.

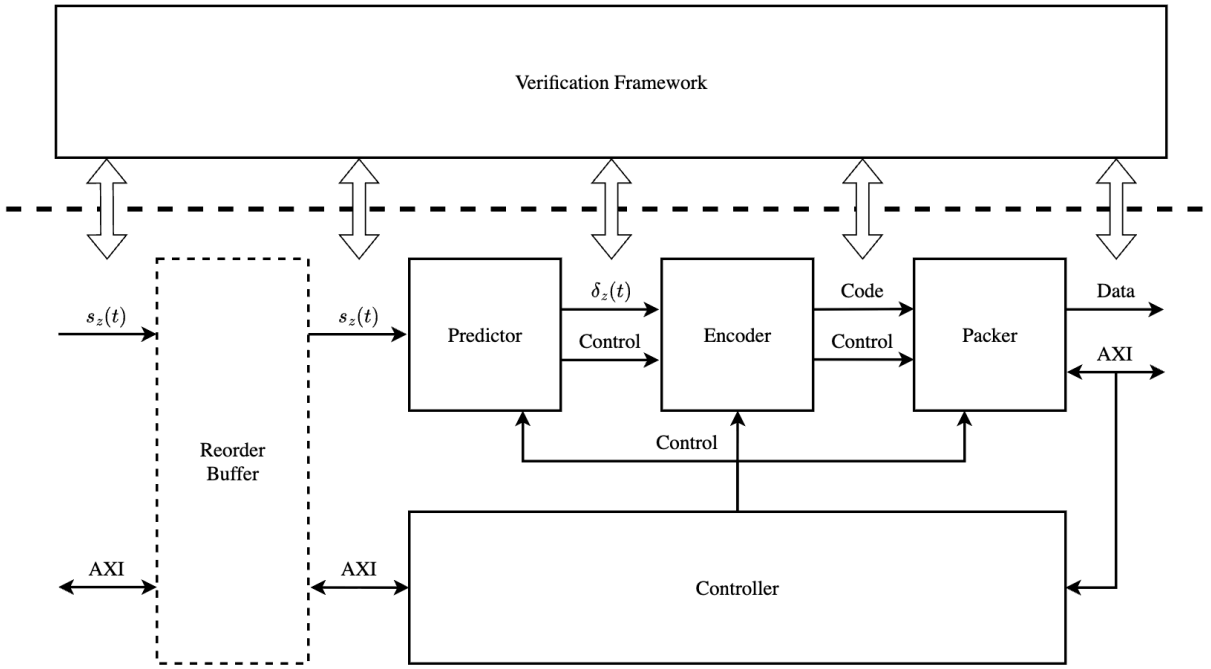


Figure 4.1: Block diagram of Vorhaug. For easy integration into HYPSO, a reorder buffer reorders samples from BIP to BIL.

The controller module keeps track of the sample index of the current input sample and passes it through a control object to the predictor module. Additionally, a global valid signal can be used to freeze the calculation if needed. The predictor and encoder modules are responsible for producing the mapped quantizer indices and compressed image bitstream, respectively. Outputs of the encoder are buffered in the packer module and shaped into blocks matching the data width of the output AXI bus. The accelerator uses a pipelined architecture to achieve high throughput. Once the control object passed to the first stage of the predictor has propagated through the pipeline and arrived at packer, the first valid block is output at the data bus. The predictor is the main focus of this work, and is covered in greater detail by the following sections.

Figure 4.2 shows the pipeline of the prediction stage of the accelerator. By use of the mathematical optimization approach, the need for results from the sample representative stage to calculate the double resolution prediction error (Equation 2.17) is eliminated. This, in combination with speculative calculation of the two possible outcomes of the sign calculation, results in a pipeline similar to that of Figure 3.3. The prediction calculation and weight updates are done in pipeline stage three. The prediction calculation, specifically the predicted central local difference (Equation 2.12), depends on the updated weight. In turn, the weight update depends on the result of the prediction calculation. Because of this, the weight update calculation cannot be split into multiple stages, limiting the achievable clock speed and throughput of the design.

Calculation of the quantizer index $q_z(t)$ and the predicted sample and sample endpoint difference $\theta_z(t)$ of equations 2.3 and 2.8 both require a division by $2m_z(t)+1$. These operations are started in parallel in stage four, and are performed using a shift/subtract algorithm [22, pp. 256–268]. The algorithm allows a pipelined divider with a length dependant on the bit depth of the operands, meaning that the pipeline depth of the predictor is dictated by the dynamic range D.

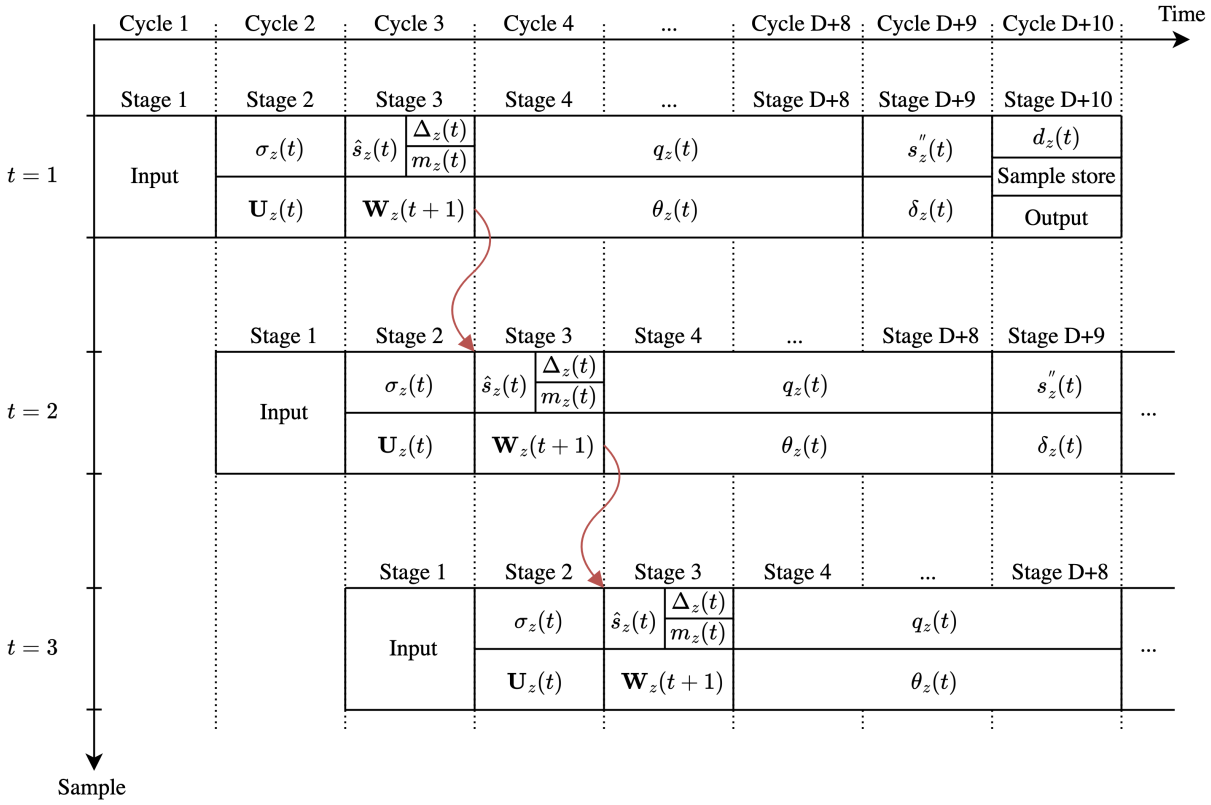


Figure 4.2: Full pipeline of Vorhaug. The indivisible weight update calculation in stage 3 restricts the throughput due to the dependency indicated by the red arrows.

In order to provide necessary values to calculations, a set of shift registers are used to buffer values until they are needed. Values are pushed into the registers when ready and read by stages at different taps. Special buffers are used to provide the sample representatives and local differences that constitute the prediction neighborhood, and are made using a set of First-In-First-Out (FIFO) buffers. See the work of Vorhaug for a full explanation.

The accelerator was verified using the verification model previously mentioned. A verification framework reads intermediary values from the design during simulation and compares them to the results of the verification model. Inputs and outputs are passed to the design using a simulated AXI interface. Test data is randomly generated, providing sufficient coverage of the different configuration of image sizes and header configurations.

For the implementation of the delayed weight updates proposed by Jia et al., the weight update calculation of stage three is the most interesting of the accelerator (Figure 4.3). The hardware shown in the figure is duplicated for each of the elements in the weight vector, indexed by i. The two possible outcomes of the weight update are calculated in parallel with the sign of the prediction error. After resolution of the prediction and sign calculations, the correct path is selected. For $t = 0$, initial weight values following the default initialization method (Equation 2.14) are selected. Each band keeps a separate weight vector which is individually updated. As described by Sánchez et al., the accelerator uses the BIL processing order, which processes frames sequentially (Figure 2.3). A FIFO of size N_x stores weights at the end of each line. At the beginning a new line, the weight stored from the previous frame of the same band is retrieved. For updates within a frame the weight is stored in a simple register.

The calculation of the sign of the double-resolution prediction error follows the mathematical optimization approach as described in Equation 3.1. The expression can be efficiently calculated by

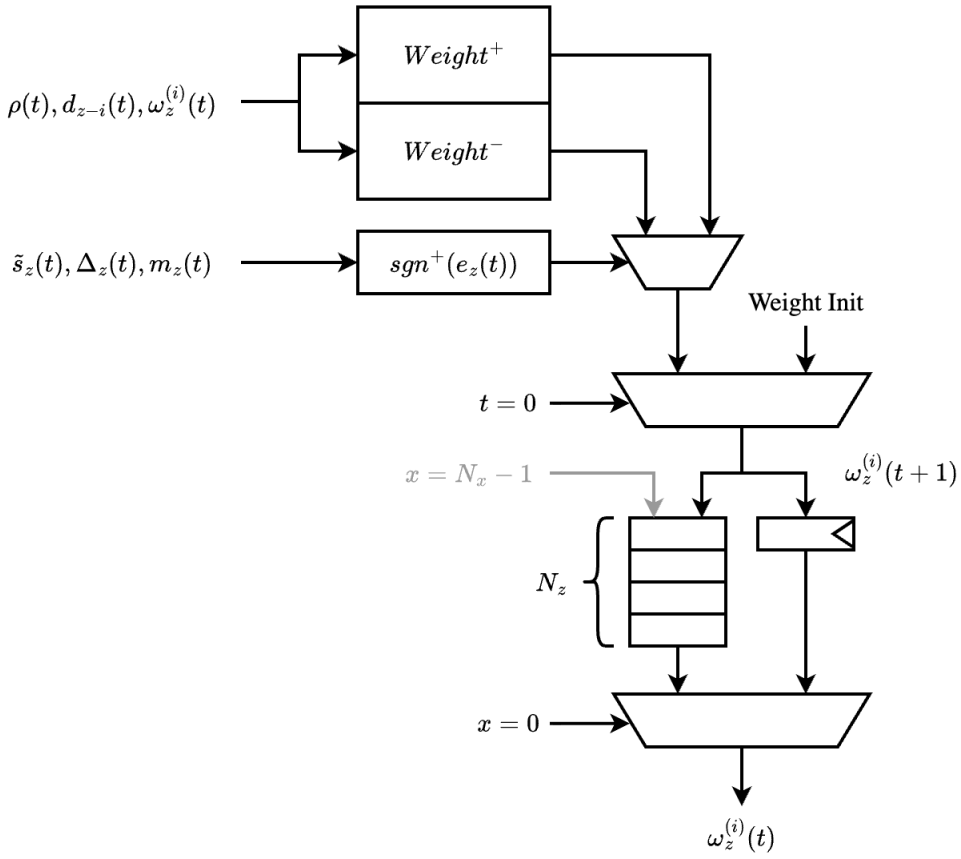


Figure 4.3: Weight update stage of Vorhaug following the mathematical optimization approach proposed by Sánchez et al. Weight updates are calculated speculatively while the sign of the prediction error resolves.

digital hardware using the following boolean expression:

$$\begin{cases} !\big(\tilde{s}_z(t)'\text{LSB}\big), & |\Delta_z(t)| \leq m_z(t), \\ !\big(\Delta_z(t)'\text{MSB}\big), & \text{otherwise.} \end{cases} \tag{4.1}$$

Here the difference between the sample and predicted sample is denoted as the prediction residual $\Delta_z(t) = s_z(t) - \hat{s}_z(t)$. Here a boolean 1 corresponds to a negative sign, and a boolean 0 to a positive sign. The Least Significant Bit (LSB) of $\tilde{s}_z(t)$ indicates whether it is odd or even, meaning that the upper clause corresponds to the two upped clauses of Equation 3.1. The MSB of $\Delta_z(t)$ gives its sign when represented using twos complement.

Chapter 5

Implementation and Verification

This section outlines the integration of the new weight update scheme proposed by Jia et al. into the hardware accelerator implemented by Vorhaug. The use of this new weight update scheme results in a compressor that is no longer compliant with the issue 2 standard. Because of this, the CNES decompressor can no longer be used to decompress, and thus verify, the results. No open-source issue 2 compliant decompressors are available, prompting the need for the development of such a tool. Before the changes to the hardware accelerator are presented, the outline for developing an open-source issue 2 decompressor is covered.

5.1 Open-source Decompressor

As shown in section 2.1, the issue 2 compressor and decompressor have many similarities. This is especially true for the predictor stage, but also applies to the processing of header configuration. To shorten the development time, the verification model written by Vorhaug was used as a starting point for the implementation of the decompressor. As opposed to the verification model, the decompressor is not only needed for verification purposes. To successfully integrate the delayed weight updates into the on-board compressor, the decompressor must be able to operate on full scale satellite images. This means that the execution time and memory usage of the program must be kept in mind during development.

Figure 5.1: Crop of HYPSO image used for profiling of the verification model, marked with white, stippled outline.

To analyse the execution time and memory usage of the verification model, profiling tools were used. These track the CPU time and memory usage of the program during execution, and the results are usually visualised using a flamegraph. The analysis done here only aims to give a rough overview of the most time and memory consuming sections of the program, and thus only a simple analysis is made. The stippled outline of the HYPSO image shown in Figure 5.1 was used as the input for the verification model during profiling. The cropped image has 80 spectral bands and a spatial size of 70x150 samples. The compressor was configured in reduced prediction mode with narrow local sums, using the hybrid encoder and three prediction bands, giving a compression ratio of 2.47.

The profiling results are summarized in Table 5.1, showing the total execution time and memory usage. Results vary depending on the characteristics of the image used, algorithm configuration, and hardware used to run the program. Thus, the numbers shown in the table should only be used as a hint as to the performance of different stages of the calculation. Most notably, the prediction stage occupies 65.2% of the execution time. Additionally, a significant amount of time is spent storing the intermediary results to files. This is because moving data from memory to disk is very slow. The hybrid encoder is responsible for 83.1% of the heap memory allocated by the program. Note that even for such a small image, a rather large chunk of memory (1203MB) is allocated. Based on this analysis, two main approaches to improve the performance of the decompressor are used: a full rewrite of the predictor stage and setting the storing of intermediate results as optional. This targets the two main time consumers of the program, namely the predictor and saving of outputs to disk.

Table 5.1: CPU time and memory use profile for image compression using verification model.

Module	**CPU Time**	**Memory Use**
Total	22.9s	1203MB
Loading	0.3%	5.7%
Predictor	65.2%	11.2%
Hybrid encoder	16.7%	83.1%
Save outputs	17.8%	N/A

Python is an interpreted language with dynamic typing. Such programming languages are known to be significantly slower than traditional compiled languages with static typing. Commonly, Python libraries such as Numpy are implemented in compiled languages such as C, C++ or Rust. By use of special libraries and builtin support in the Python language, bindings to the compiled library are created that allow classes and other objects to be used by Python. This combines the simple syntax and fast development times enabled by Python with the high speeds of precompiled libraries. In order to improve the performance of the predictor stage while reusing the encoder and header readers of the old verification model, this work rewrites the predictor stage in C++. By use of the library *pybind11*, bindings are created that allow the new predictor class to be instantiated by the old Python framework. The interface of the new class is identical to that of the old class, while allowing data to be passed between the C++ and Python programs at runtime.

5.2 Hardware Modifications

5.3 Hardware Verification

5.4 Hardware Testing

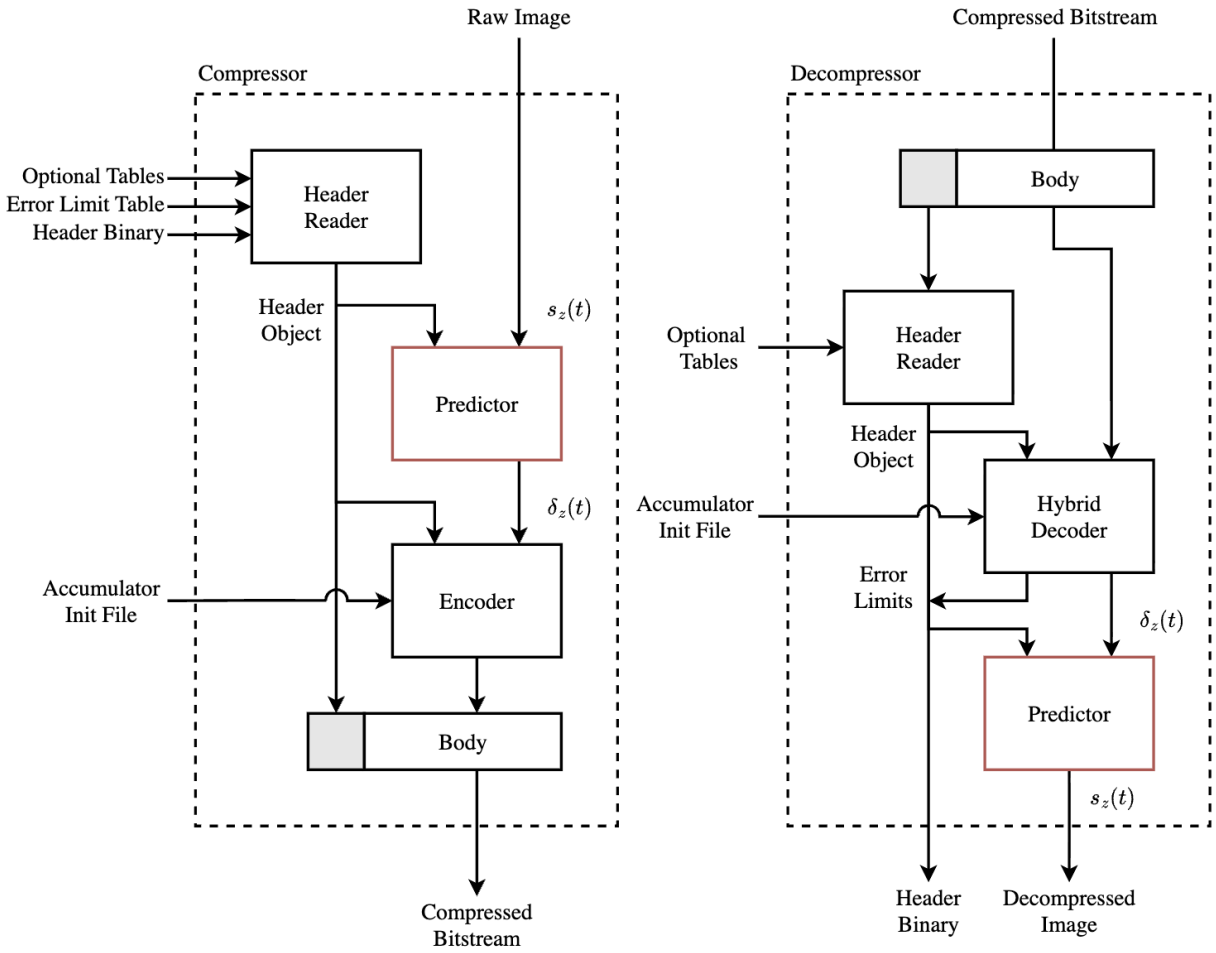

Raw Image
Compressor
Optional Tables
Error Limit Table
Header Binary
Header Reader
Header Object
$s_z(t)$
Predictor
$\delta_z(t)$
Accumulator Init File
Encoder
Body
Compressed Bitstream
Compressed Bitstream
Decompressor
Body
Optional Tables
Header Reader
Header Object
Accumulator Init File
Hybrid Decoder
Error Limits
$\delta_z(t)$
Predictor
$s_z(t)$
Header Binary
Decompressed Image

Figure 5.2

Code listing 5.1: Pseudocode for hybrid decoder.

```
accumulator, active_prefixes = decode_image_tail()
counter                      = get_counter_value(t_max)

for all z and t: # iterate in reverse order
    mapped_quantizer_index[z, t] = decode_sample(z, t)
    if ordering = BI and prediodic_error_limit_updating:
        header.error_limits[z] = decode_error_limts(z)

def decode_sample(z, t): # pops bits from compressed bitstream
    if t = 0: # first sample encoded as is
        mapped_quantizer_index[z, t] = read_D_bit_integer()
    else:
        if accumulator[z, t] * 2^14 >= threshold[0] * counter[t]:
            mapped_quantizer_index[z, t] = decode_high_entropy(z, t)
        else:
            mapped_quantizer_index[z, t] = decode_low_entropy(z, t)

    t_next = t-1
    counter[t_next] = get_counter_value(t_next)

    # update accumulator
    if counter[t_next] = 2^rescaling_counter_size - 1: # rescaling
        rounding_bit = bitstream.pop()
        updated_accumulator = 2 * accumulator[z, t] - 4 * mapped_quantizer_index[z, t] - 1

        if rounding_bit = 0 and updated_accumulator_lsb = 1:
            updated_accumulator += 1 # ensure even
        else if rounding_bit = 1 and updated_accumulator_lsb = 0:
            updated_accumulator += 1 # ensure odd
        accumulator[z, t_next] = updated_accumulator

    else: # normal case
        accumulator[z, t_next] = accumulator[z, t] - 4 * mapped_quantizer_index[z, t]
```

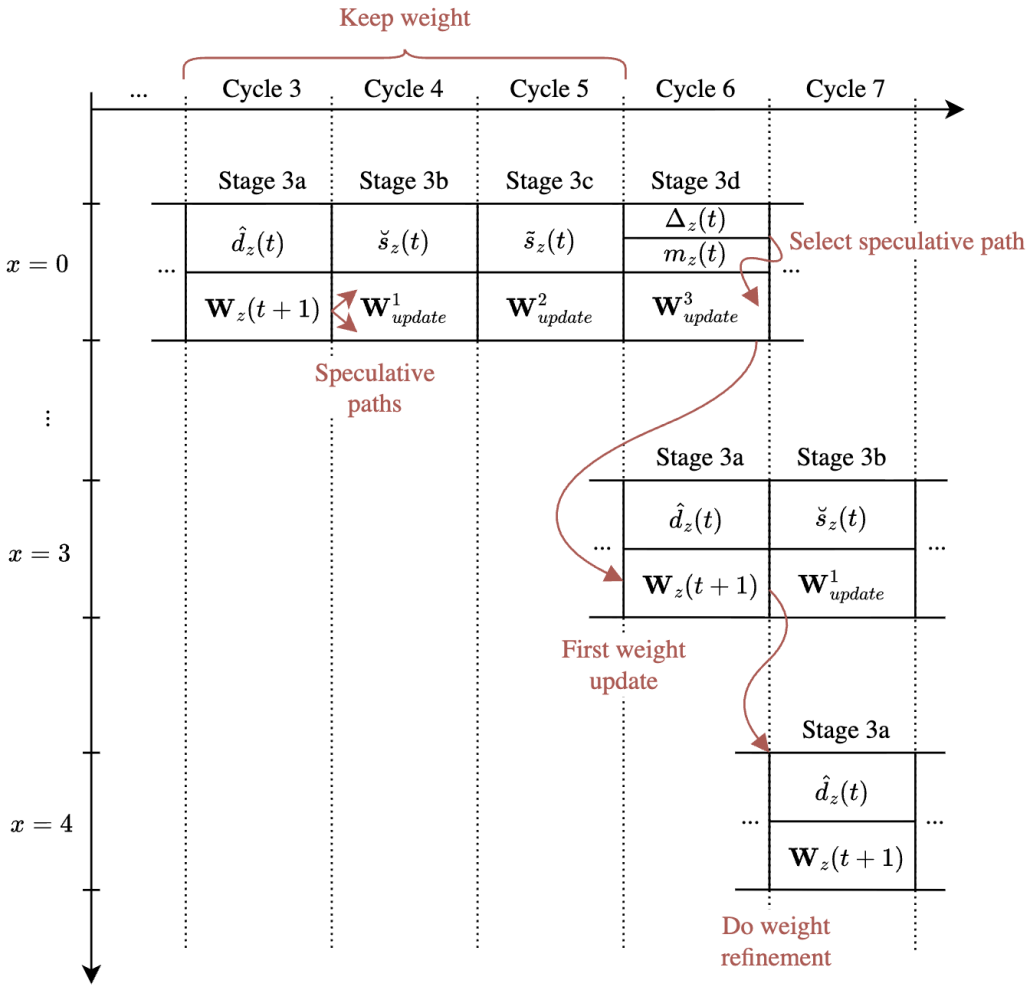


Figure 5.3: Proposed pipeline for delayed weight updates.

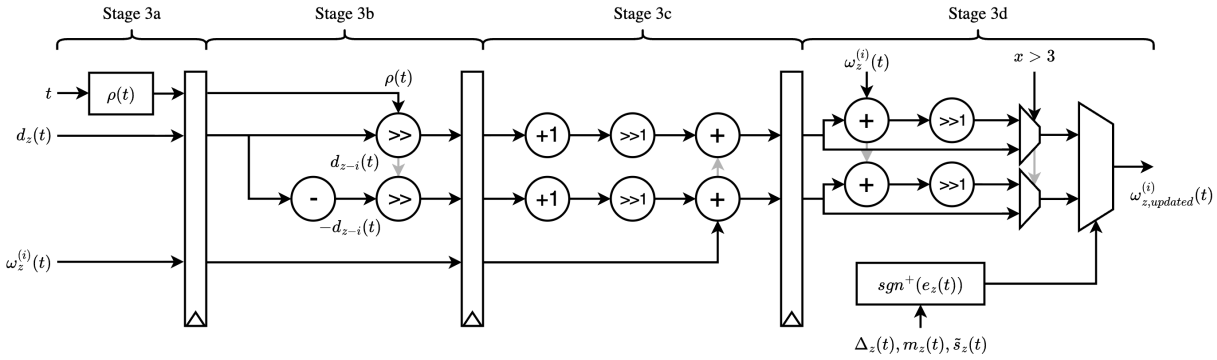


Figure 5.4

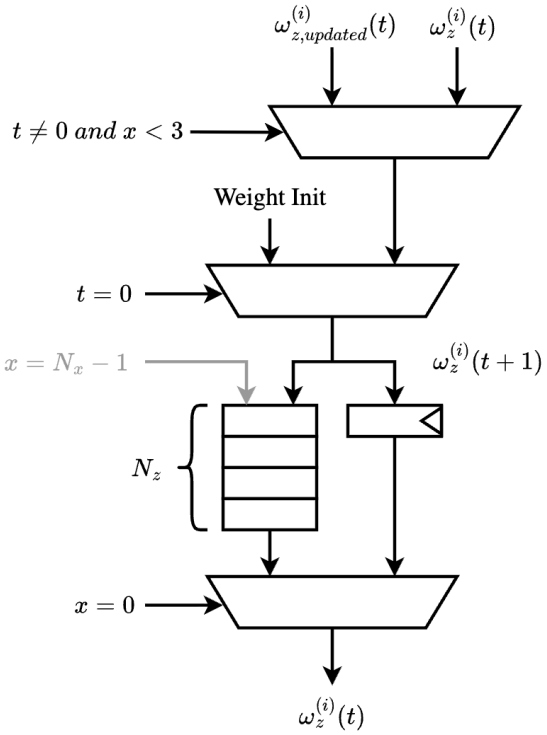


Figure 5.5

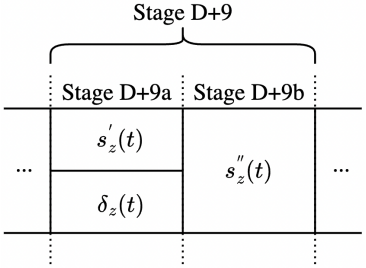


Figure 5.6: Sample representative stage split into two stages $D+9a$ and $D+9b$, reducing the new critical path after splitting the weight update stage.

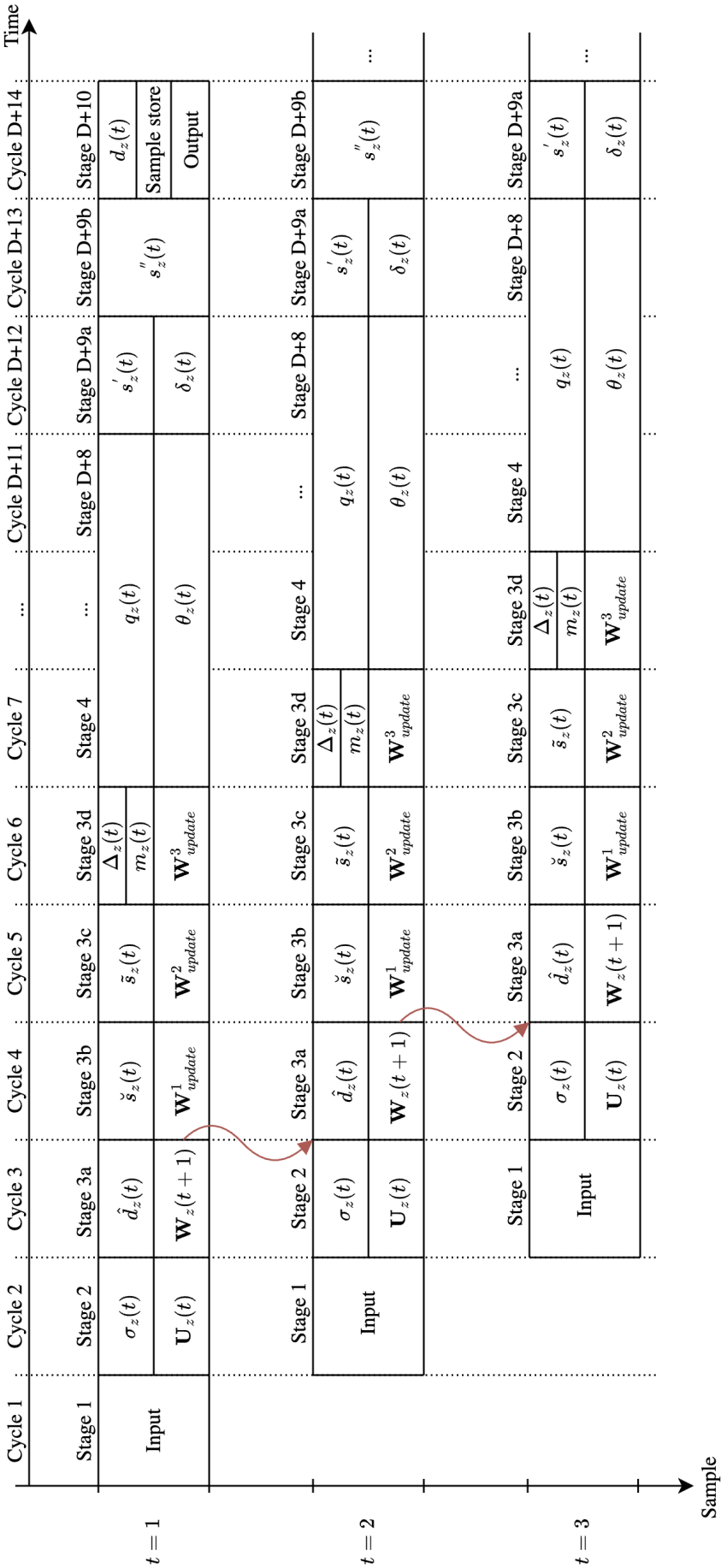


Figure 5.7: Proposed pipeline with delayed weight updates allowing splitting stage 3 into four separate stages, 3a to 3d. Additionally, the sample representative calculation in stage $D+9$ is split into two stages $D+9a$ and $D+9b$.

Table 5.2: Selected predictor and hybrid encoder parameters. Values are selected based on the work by Blanes et al., and are the same as those used by Vorhaug [5, 23]. Table recreated from [5].

Name	**Symbol**	**Configuration**
Sample type		Unsigned
Dynamic range	D	16
Sample encoding order		BIL
Entropy coder type		Hybrid coder
Quantizer method		Absolute
Number of prediction bands	P	3
Register size	R	64
Local sum type		Narrow column-oriented
Prediction mode		Reduced
Weight component resolution	Ω	19
Weight initialization method		Default
Weight update initial parameter	$\nu_{\min}$	-1
Weight update final parameter	$\nu_{\max}$	4
Weight update change interval	t_{inc}	2^6
Weight update exponent offsets	$\zeta_z^{(i)}, \zeta_z^*$	0
Periodic error limit update		Disabled
Band-varying absolute limits		Disabled
Absolute error limit	a_z	*Varied*
Band-varying relative limits		Disabled
Relative error limit	r_z	0
Sample representative resolution	Θ	3
Sample representative offset	ψ_z	7
Band-varying offset		Disabled
Sample representative damping	θ_z	3
Band-varying damping		Disabled
Unary length limit	$U_{\max}$	18
Rescaling counter size	γ^*	6
Initial count exponent	γ_0	1
Accumulator initialization value	$\tilde{\Sigma}_z(0)$	$4 \cdot 2^{\gamma_0}$

Chapter 6

Results

6.1 Verification Model

Table 6.1: CPU time and memory use profile for image compression using C++ predictor and reduced memory usage.

Module	CPU Time	Memory Use
Total	9.4s	438MB
Loading	0.9%	17.2%
Predictor	72.6%	13.8%
Hybrid encoder	26.4%	69.0%
Save outputs	<0.1%	N/A

Table 6.2: Compression performance, execution time of predictor in parenthesis (TODO: find better way of displaying than in parenthesis)

Image	Predictor	Samples	Time (s)		Throughput (MS/s)	Memory (MB)
Hypso 2	C++	78361920	821	(293)	0.095	32415
Aviris	C++	77643776	1166	(290)	0.066	32091
Landsat	C++	6291456	103	(22)	0.061	2669
Hypso 2	Python	78361920	2818	(2169)	0.027	38996
Landsat	Python	6291456	263	(161)	0.024	3204

Table 6.3: Decompression

Image	Samples	Time (s)	Throughput (MS/s)	Memory (MB)
Hypso 2	78361920	1093	0.072	32414
Aviris	77643776	1834	0.042	32091
Landsat	6291456	149	0.042	2667

6.1.1 Debugging

Clamping of reconstructed sample.

Selection of quantization bin center.

Rounding of signed integers in python vs C++.

6.2 Compression Performance

Table 6.4: Delayed weight updates, relative 16 and absolute 4

Image	**Delayed**	**Ordinary**	**Decrease**
Hypso 2	2.47	2.49	−0.80%
Aviris	4.31	4.41	−2.23%
Landsat	4.82	5.21	−7.49%

The difference in compression performance shown in Table 6.4, especially that of Landsat, are likely due to the characteristics of the specific images used. For example, the Landsat image only has six spectral bands, as opposed to 120 and 200 (?) for Hypso and Aviris.

Table 6.5: Reductions in compression performance for coastline image for this work and the work by Jia et al.

PAE	Coastline, Jia et al.	Coastline, this work
0	−1.53%	−0.50%
1	−2.88%	−0.78%
3	−3.86%	−0.63%
7	−4.30%	−0.98%
15	−1.23%	−1.10%

Table 6.6: PSNR of decompressed images with delayed weight updates. Difference compared to without delayed updates are negligible, exact same when rounded to two decimals.

PAE	CR	PSNR (dB)
0	2.02	N/A
1	2.53	225.86
3	3.15	207.94
7	4.03	192.54
15	5.39	178.10

Note that the delayed weight update scheme does not affect the PSNR of the reconstructed images. This is due to the maximum error feature of the algorithm. Only the compression performance is affected.

Figure 6.1: Decoded Hypso 2 image

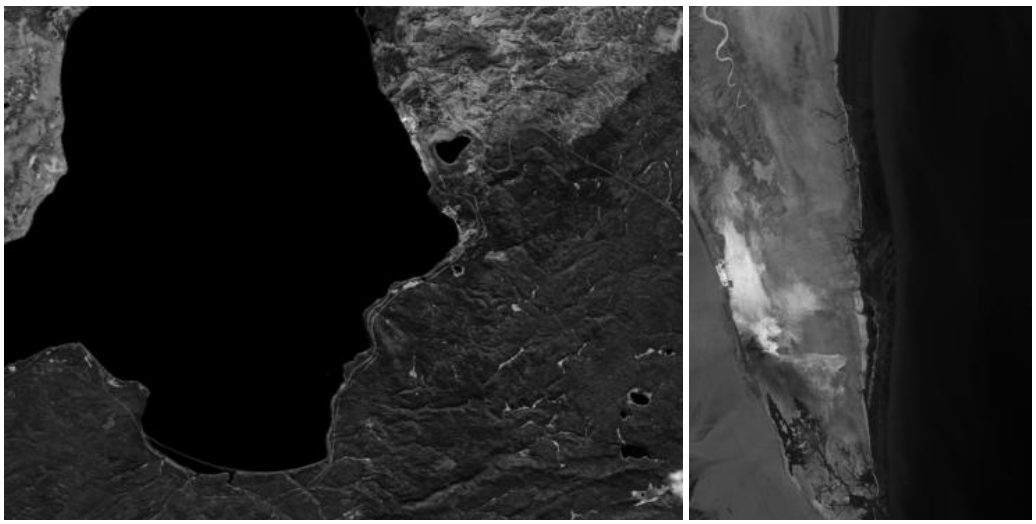


Figure 6.2: Decoded Aviris and Landsat images.

6.3 RTL Results

6.4 Hardware Testing

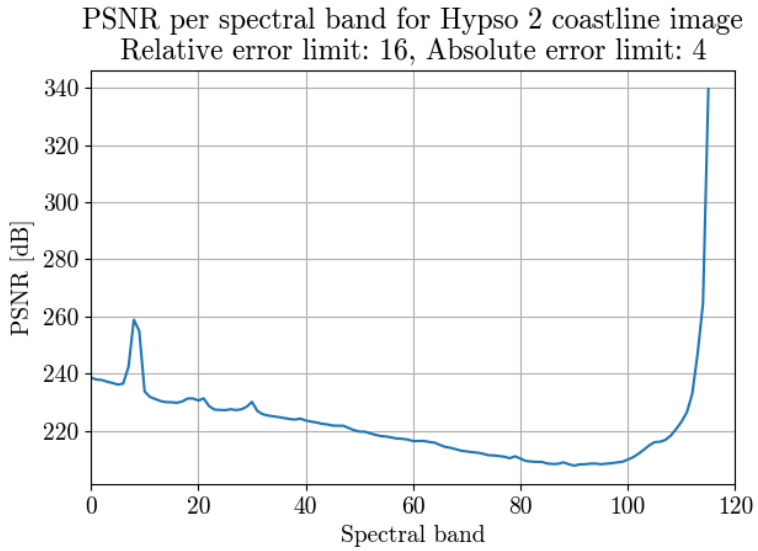


Figure 6.3

Table 6.7

FPGA	LUT	FF	DSP	BRAM	Power	Frequency
Zynq-7030	5958	3425	4	83.5	152 mW	90.9 MHz
Zynq ZU7EV	6331	3442	4	79.5	273 mW	200.0 MHz
Zynq-7030 (predictor)	2841	2212	4	69.5	263 mW	125.0 MHz
Zynq ZU7EV (predictor)	2969	2185	4	65.5	467 mW	312.5 MHz

Table 6.8: Throughputs of hardware implementations of the issue 2 algorithm in open literature.

Implementation	FPGA	LUT	FF	DSP	BRAM	Throughput
Barrios et al. [17]	XCKU040	17185	11915	63	85	18 MS/s
Sánchez et al. [18, 19], predictor	XCKU040	19840	12970	33	213	231 MS/s
Barrios et al. [19], predictor	XCKU040	11095	11729	20	186.5	236 MS/s
Báscones et al. [20]	XQRKU060	16458	15707	30	187.5	285 MS/s
Chatziantoniou et al. [21] 1 core	XCKU040	18861	21395	-	104	285 MS/s
Chatziantoniou et al. [21] 5 core	XCKU040	67362	68987	-	306	1375 MS/s
Vorhaug et al. [5]	XCKU040	5577	2635	4	57.5	110 MS/s
Vorhaug et al. [5]	XCZU7EV	5475	2635	4	57.5	141 MS/s
This work	XCZU7EV	6331	3442	4	79.5	200 MS/s
Jia et al. [4], predictor	XCZU5EV	3995	5050	7	94	348 MS/s
This work, predictor	XCZU7EV	2969	2185	4	65.5	313 MS/s

Chapter 7

Discussion and Further Work

Work should be done to properly integrate the issue 2 implementation into the HYPSO-2 on-board-processing (OPU) system. This includes writing software for header, and selection of parameters. As of now, HYPSO-2 still uses the outdated issue 1 implementation. Although fast, it cannot achieve compression rates close to those of issue 2. Issue 2 for HYPSO-3!! Header software should be written in C, for easy integration with the existing software.

The decrease in compression performance introduced by delayed weight updates cause significantly longer downlink time which is not outweighed by the reduced execution time, by far.

If this is to be used onboard the satellite, we need to write a faster decompressor. Two problems: serialized processing and excessive storing of intermediates. Use buffers to store only the data that is needed, similar to the hardware. Would also speed up the implementation due to simpler memory accesses.

Chapter 8

Conclusions

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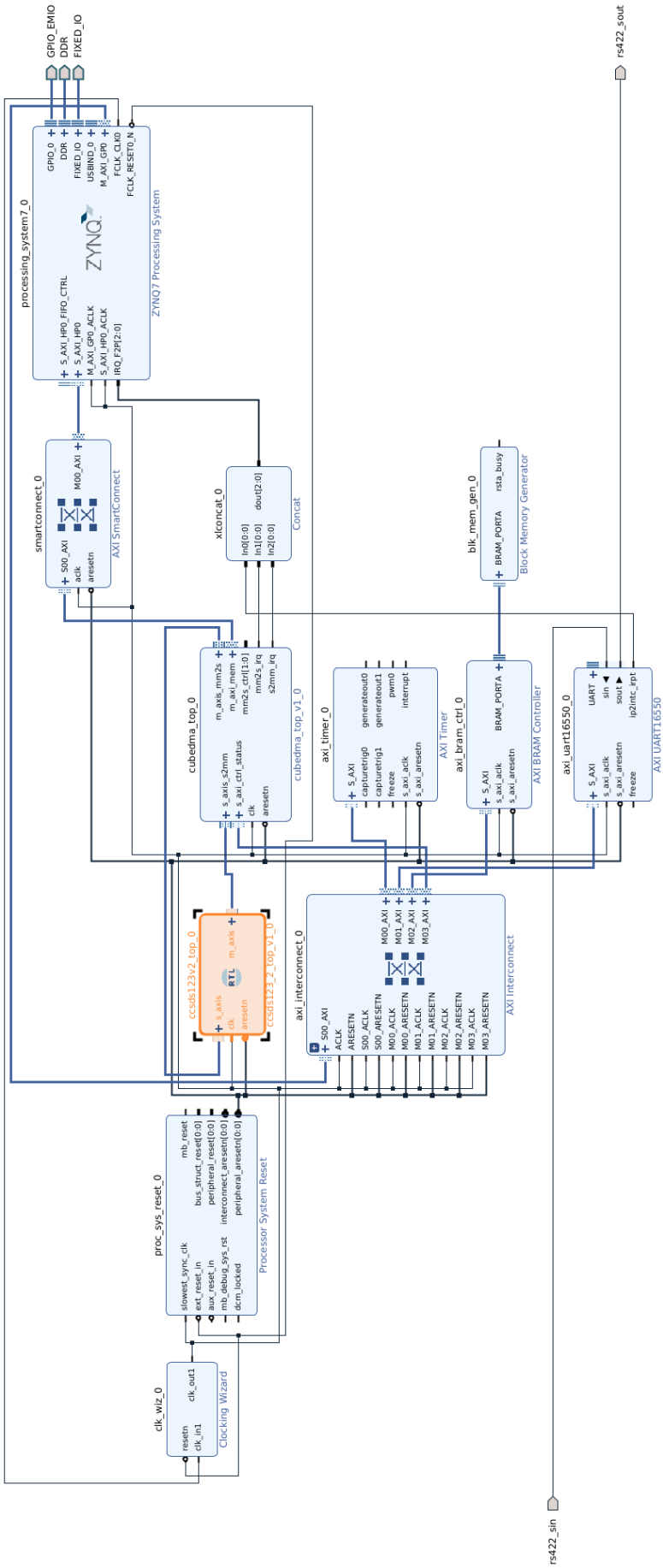
Appendix A

Block Design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DDR
FIXED_IO
rs422_sout
processing_system7_0
ZYNQ7 Processing System
smartconnect_0
AXI SmartConnect
xlconcat_0
Concat
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Block Memory Generator
cubedma_top_0
cubedma_top_v1_0
axi_timer_0
AXI Timer
axi_bram_ctrl_0
AXI BRAM Controller
axi_uart16550_0
AXI UART16550
ccsds123v2_top_0
ccsds123_2_top_v1_0
axi_interconnect_0
AXI Interconnect
proc_sys_reset_0
Processor System Reset
clk_wiz_0
Clocking Wizard
rs422_sin

Figure A.1